
UNIVERSALITY

UNIVERSAL SET

Is it possible to implement ALL the possible Boolean functions using NOT, AND, OR, NAND, NOR? Yes!

UNIVERSAL SET

What if we are not given some!

What if some are very costly! E.g., NOT

Can we reduce this set? E.g., building NOT by NAND/ NOR

SET	UNIVERSAL SET
{NOT, AND, OR, NAND, NOR}	Yes! (a Full Set)
{NOT, AND, OR}	Yes, (a Full Set) NAND = AND-NOT NOR = OR-NOT
{NOT, AND}	If we could design OR
{NOT, OR}	If we could design AND
{NOT}	If we could design AND, OR
{AND}	If we could design NOT, OR
{OR}	If we could design NOT, AND
{NAND}	If we could design NOT, AND, OR
{NOR}	If we could design NOT, AND, OR

SET	UNIVERSAL SET
{NOT, AND, OR, NAND, NOR}	Yes! (a Full Set)
{NOT, AND, OR}	Yes, (a Full Set) NAND = AND-NOT NOR = OR-NOT
{NOT, AND}	If we could design OR
{NOT, OR}	If we could design AND
{NOT}	If we could design AND, OR
{AND}	If we could design NOT, OR
{OR}	If we could design NOT, AND
{NAND}	If we could design NOT, AND, OR
{NOR}	If we could design NOT, AND, OR

UNIVERSAL SET

{NOT, AND}



Augustus De Morgan
(1806–1871)

Mathematician
Logician

DE MORGAN'S LAWS

► $(Y + X)' = Y'X'$



Augustus De Morgan
(1806–1871)

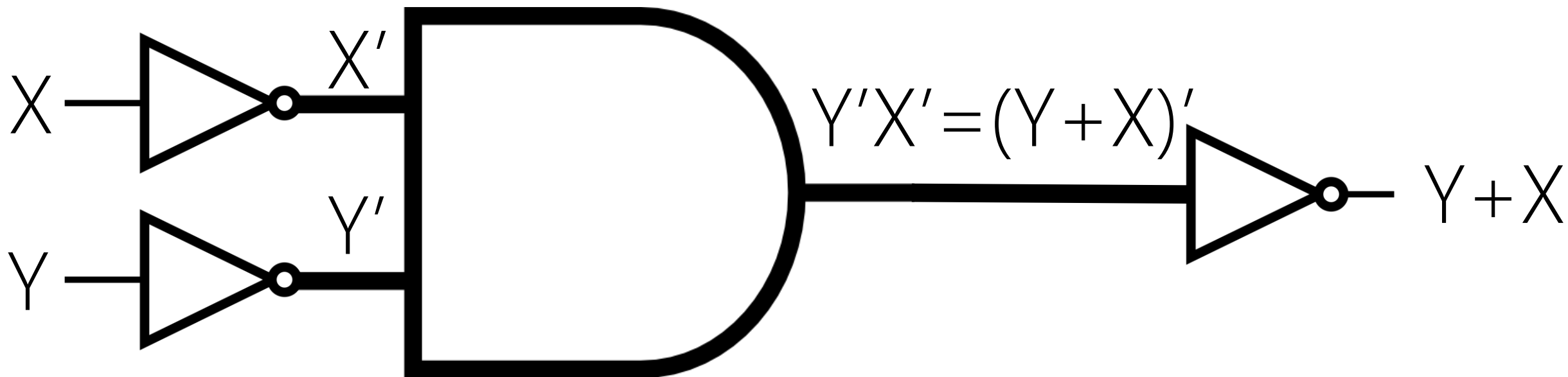
Mathematician
Logician

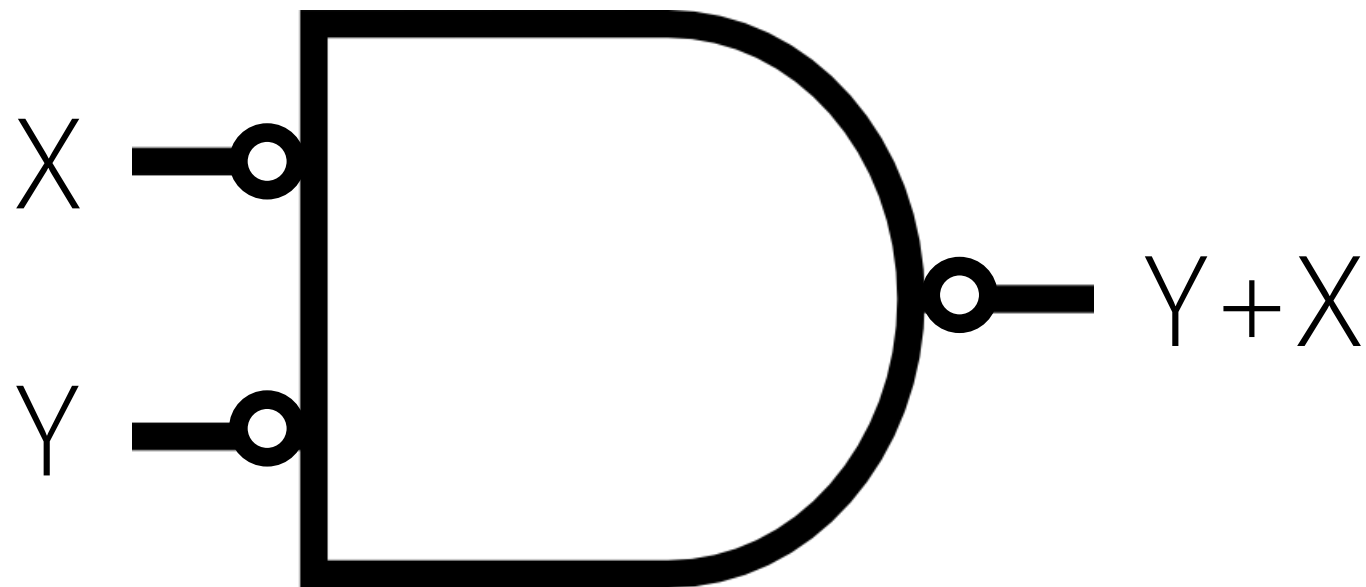
DE MORGAN'S LAWS

► $(Y + X)' = Y'X'$

$$((Y + X)')' = (Y'X')'$$

$$Y + X = (Y'X')'$$





OR

SET	UNIVERSAL SET
{NOT, AND, OR, NAND, NOR}	Yes! (a Full Set)
{NOT, AND, OR}	Yes, (a Full Set) NAND = AND-NOT NOR = OR-NOT
{NOT, AND}	If we could design OR (a Complete Set)
{NOT, OR}	If we could design AND
{NOT}	If we could design AND, OR
{AND}	If we could design NOT, OR
{OR}	If we could design NOT, AND
{NAND}	If we could design NOT, AND, OR
{NOR}	If we could design NOT, AND, OR

UNIVERSAL SET
{NOT, OR}



Augustus De Morgan
(1806–1871)

Mathematician
Logician

DE MORGAN'S LAWS

► $Y' + X' = (YX)'$



Augustus De Morgan
(1806–1871)

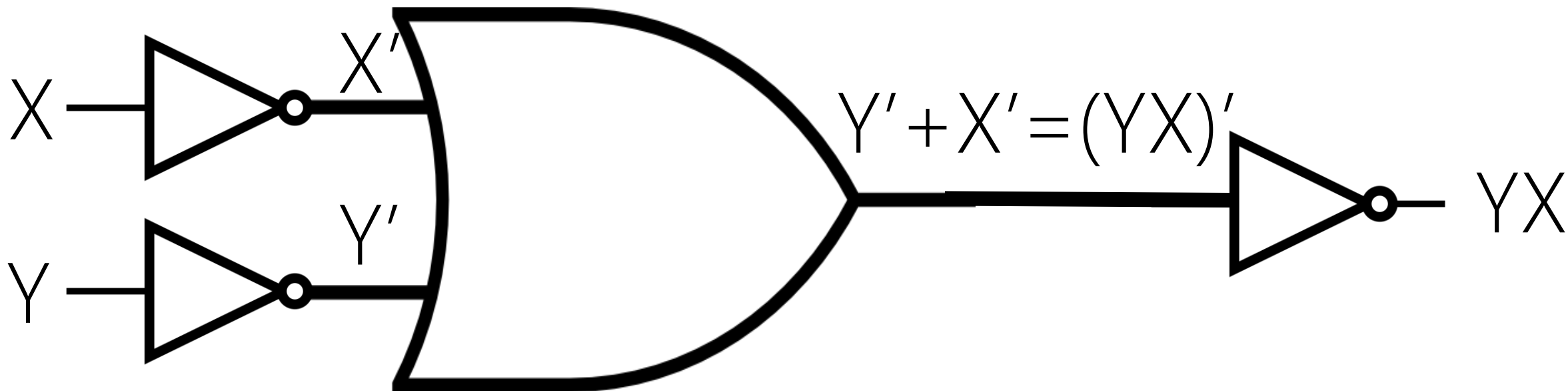
Mathematician
Logician

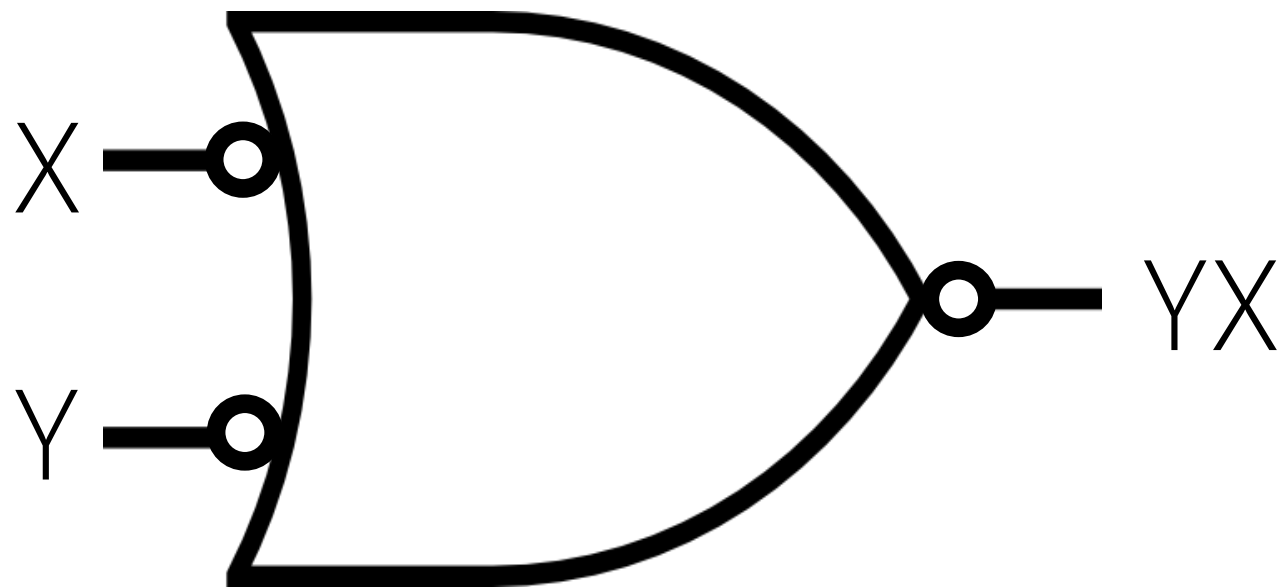
DE MORGAN'S LAWS

► $Y' + X' = (YX)'$

$$(Y' + X')' = ((YX)')'$$

$$(Y' + X')' = YX$$

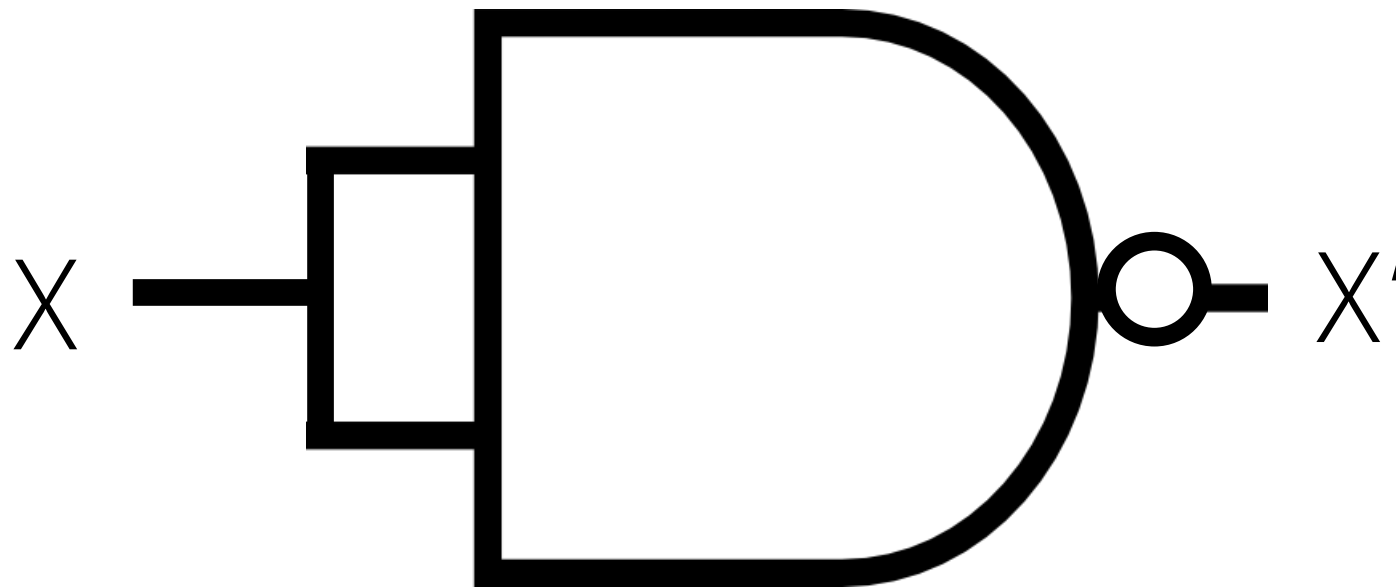




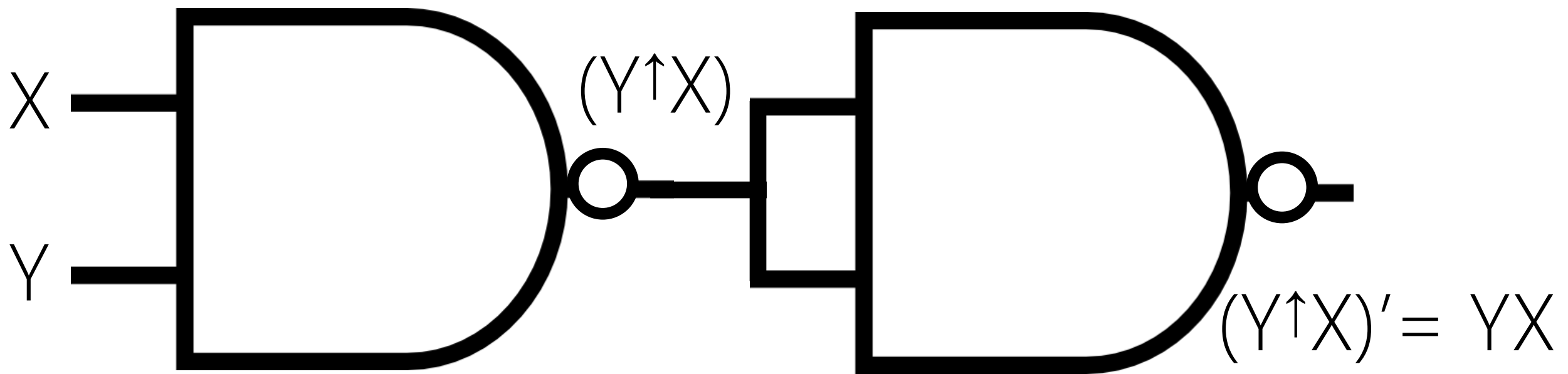
AND

SET	UNIVERSAL SET
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{NOT, OR}	If we could design AND (a Complete Set)
{NOT}	If we could design AND, OR
{AND}	If we could design NOT, OR
{OR}	If we could design NOT, AND
{NAND}	If we could design NOT, AND, OR
{NOR}	If we could design NOT, AND, OR

NOT ► $(XX)' = (X \uparrow X) = X'$



$$\text{AND} \blacktriangleright \text{NOT (NAND)} = ((Y \uparrow X))' = YX$$



OR ► DE MORGAN'S LAW

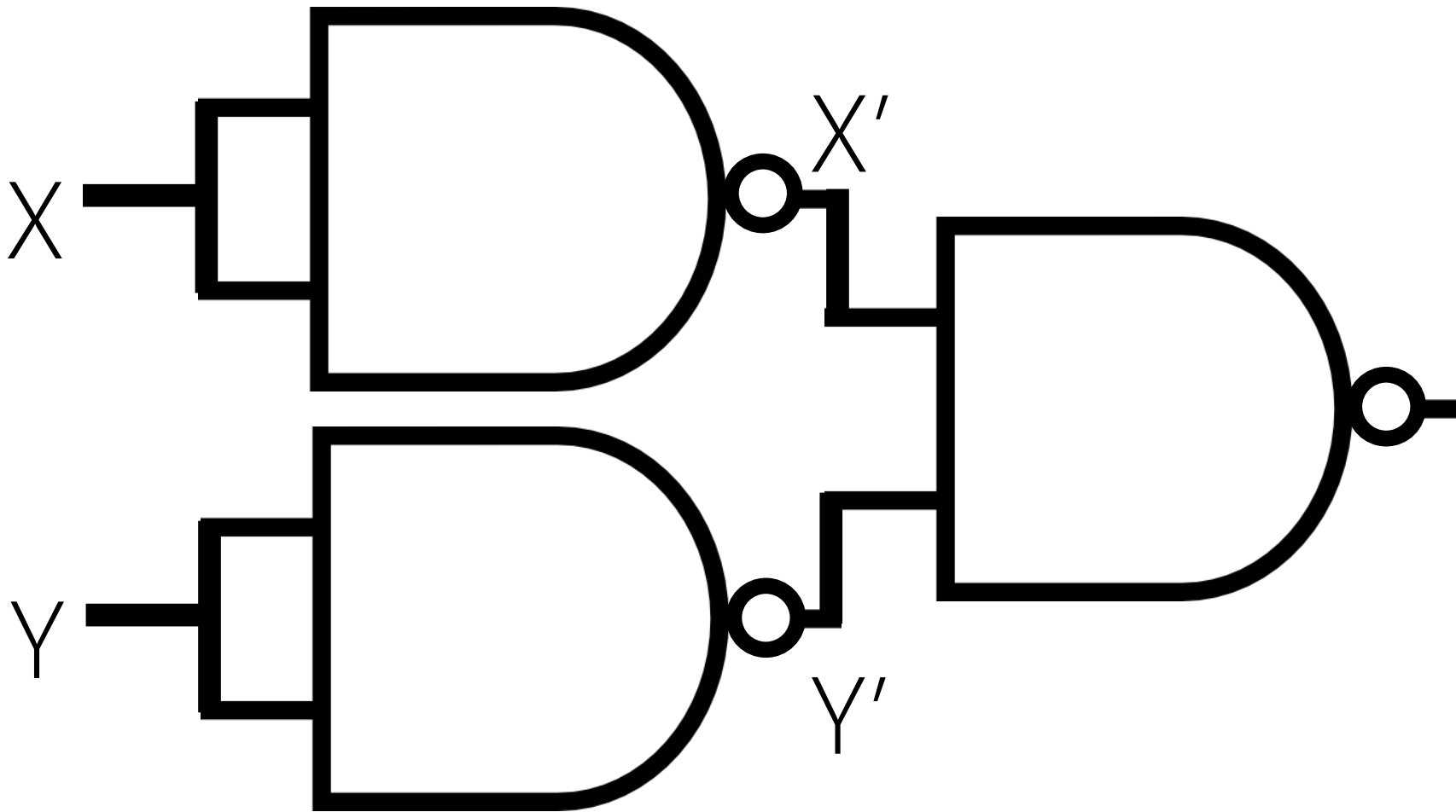
$$(Y + X)' = Y'X'$$

$$((Y + X)')' = (Y'X')'$$

$$Y + X = (Y'X')'$$

$$Y + X = Y' \uparrow X'$$

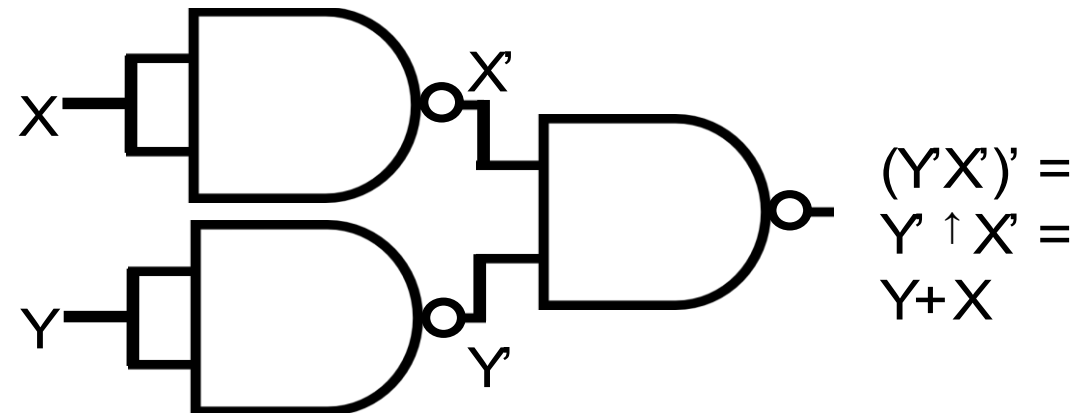
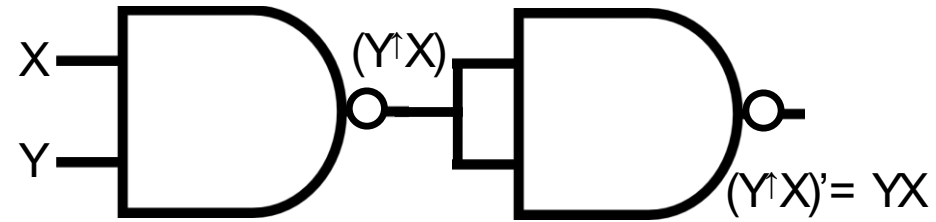
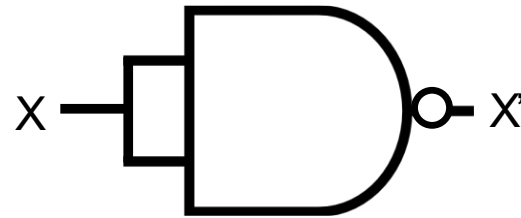
OR: DE MORGAN'S LAW



$$\begin{aligned}(Y'X')' &= \\ Y' \uparrow X' &= \\ Y + X\end{aligned}$$

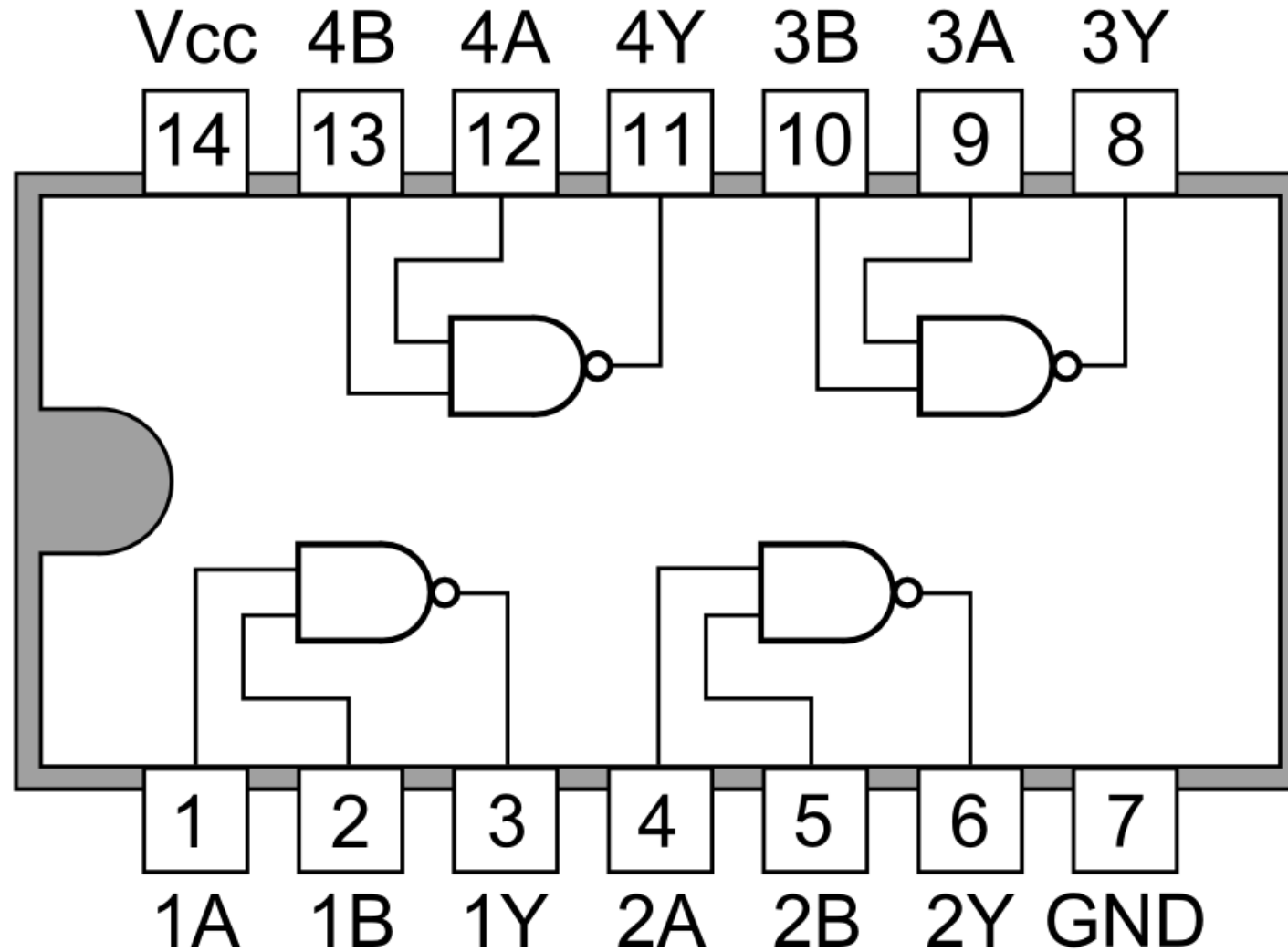
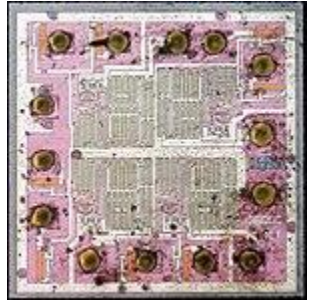
UNIVERSAL GATE

{NAND}



7400 Quad 2-input NAND Gates

https://commons.wikimedia.org/wiki/7400_series_overview



Tools & Home Improvement › Power & Hand Tools › Hand Tools › Cutters › Pipe Cutters



(5PCS) SN7400N IC Quad 2IN POS NAND GATE 14DIP 7400 SN7400

Brand: SICSTOCK

[Search this page](#)

\$**17**⁹⁹ (\$3.60 / Item)

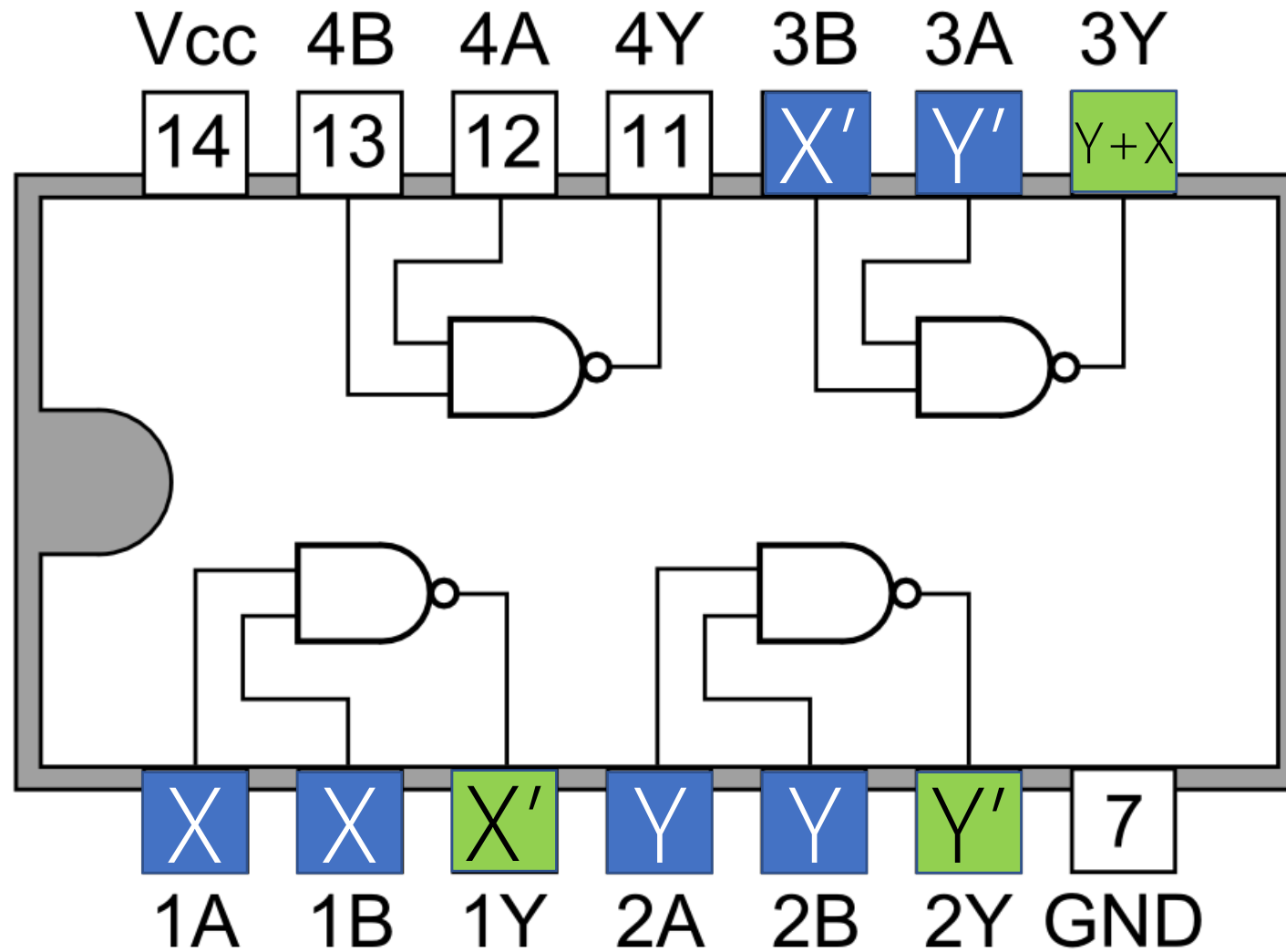
Get up to **\$150 in rewards** with the Amazon.ca Rewards Mastercard. **No annual fee.**

- Part NO.:SN7400N
- Package:14-PDIP
- Description:IC QUAD 2IN POS NAND GATE 14DIP

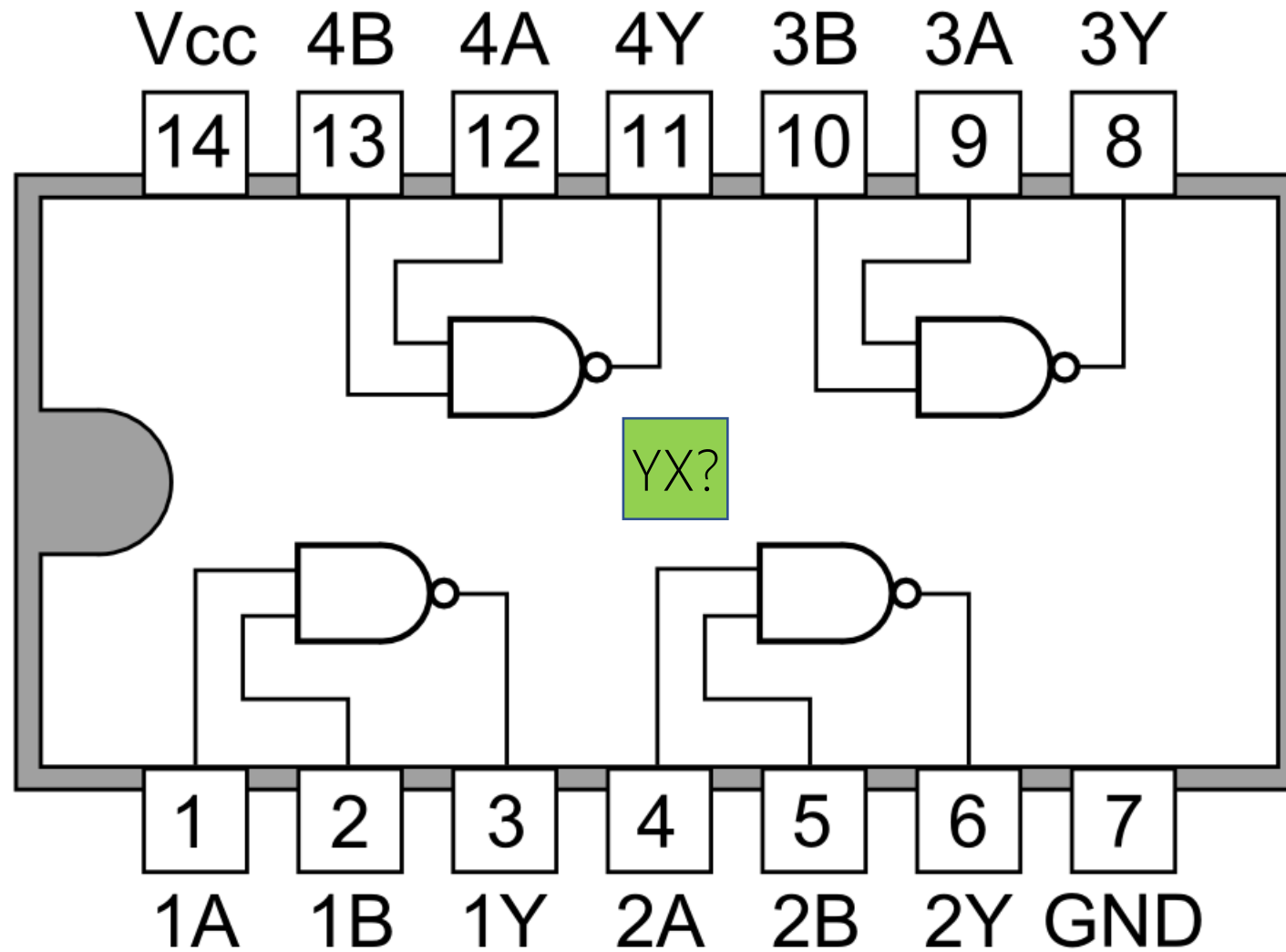


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7400 Quad 2-input NAND Gates



7400 Quad 2-input NAND Gates



SET	UNIVERSAL SET
{NOT, AND, OR, NAND, NOR}	Yes! (a Full Set)
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{OR}	If we could design NOT, AND
{NAND}	If we could design NOT, AND, OR
{NOR}	If we could design NOT, AND, OR

UNIVERSAL GATE

{NOR}

NOT ► $(X+X)' = (XX)' = X'$

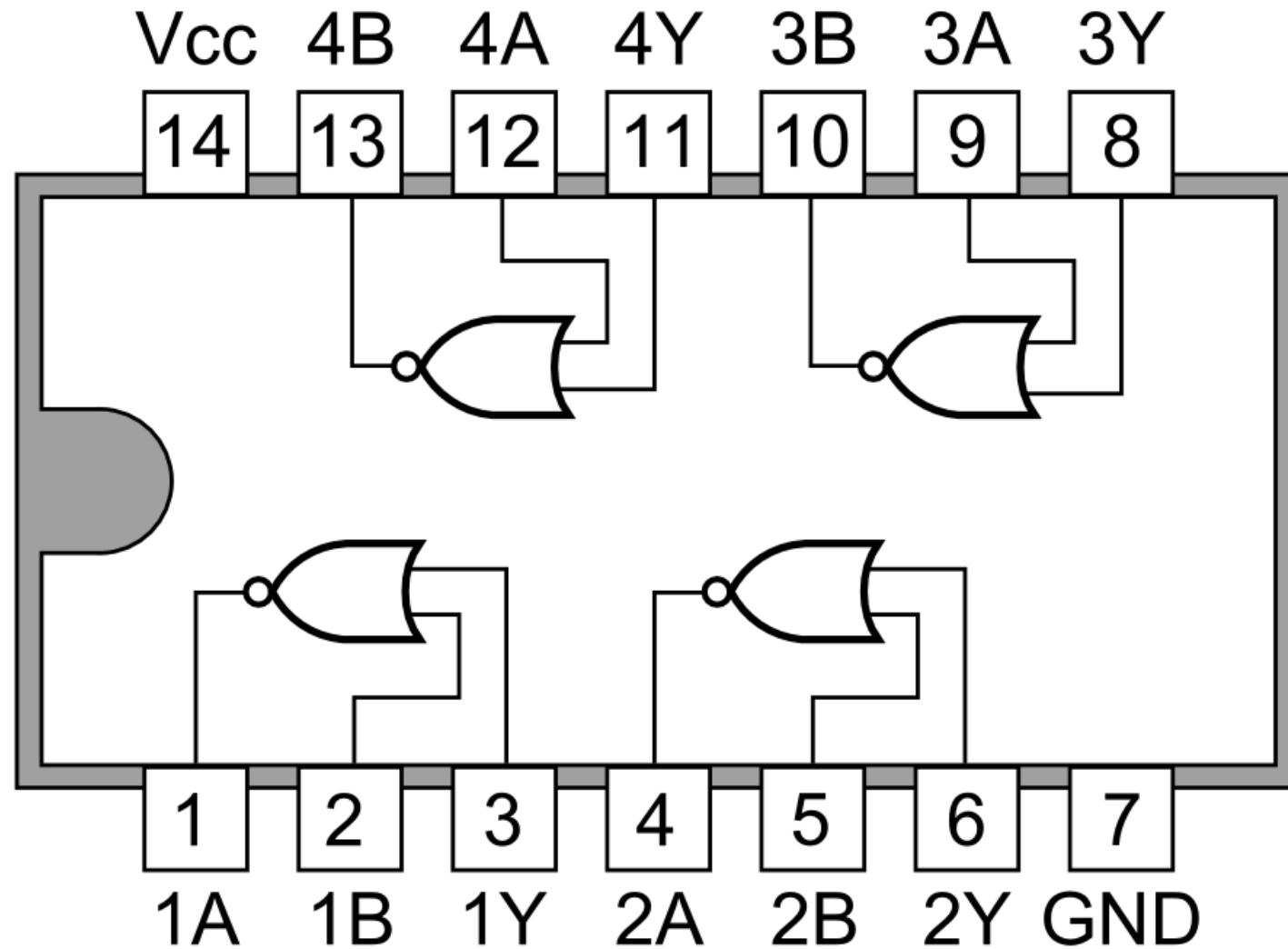
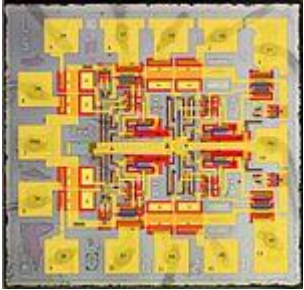
OR ► NOT (NOR)

AND ► DE MORGAN'S LAW

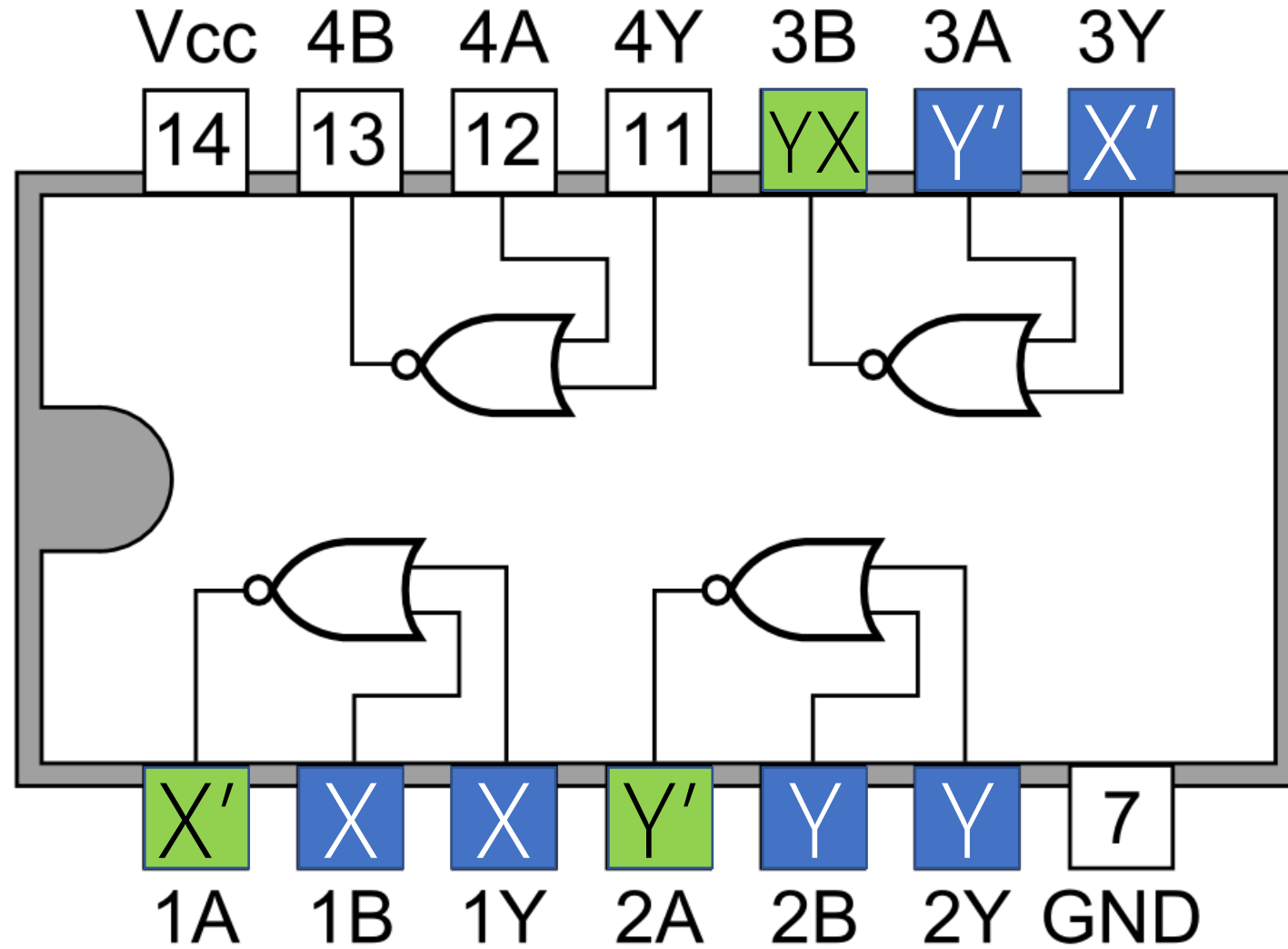
$$(Y'+X')' = YX = (Y'\downarrow X')$$

7402 Quad 2-input NOR Gates

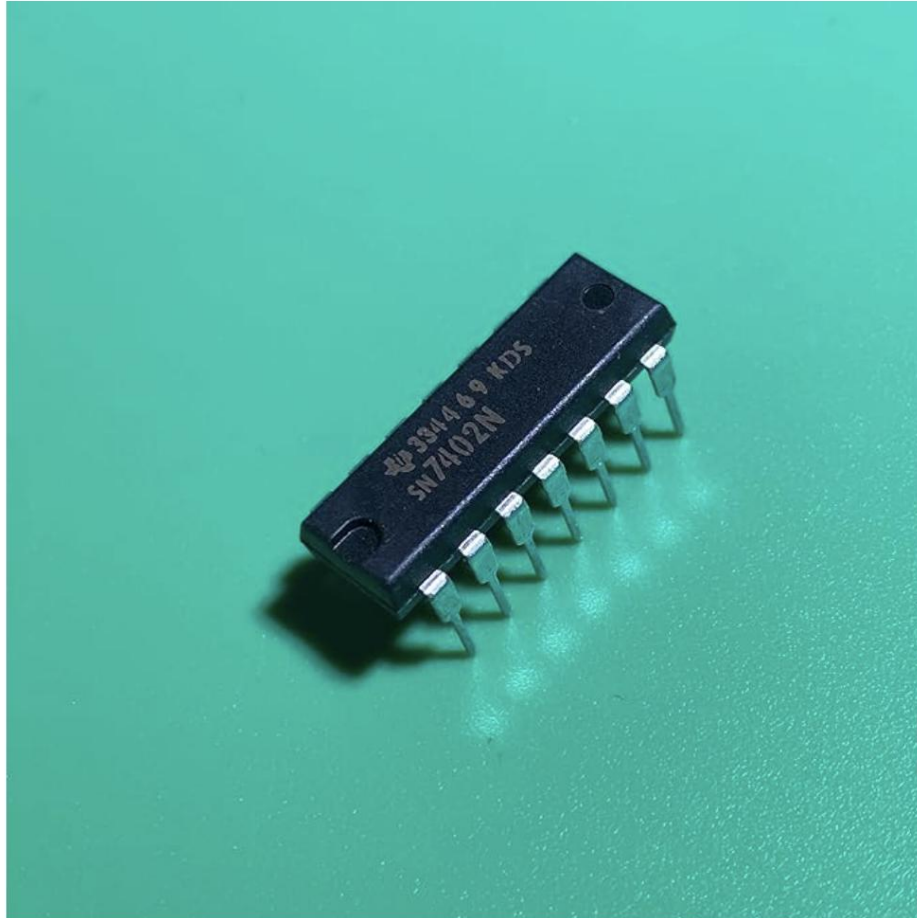
https://commons.wikimedia.org/wiki/7400_series_overview



7402 Quad 2-input NOR Gates



[◀ Back to results](#)



Roll over image to zoom in



5pcs/lot SN7402N DIP14 SN 7402N IC GATE NOR 4CH 2-INP 14DIP Quadruple 2-Input Positive- NOR Gates SN7402 N 7402 SN7402NE4 7402NG4

Brand: Generic

\$19¹⁹

Get up to **\$150 in rewards** with the Amazon.ca Rewards Mastercard. **No annual fee.**

- Power supply: AC
- Model number: SN7402NG4 SN7402AN SN7402BN S7402N S7402AN S7402BN S7402 7402AN 7402BN
- Motor type: AC Motor
- Origin: Mainland China
- Certification: NONE

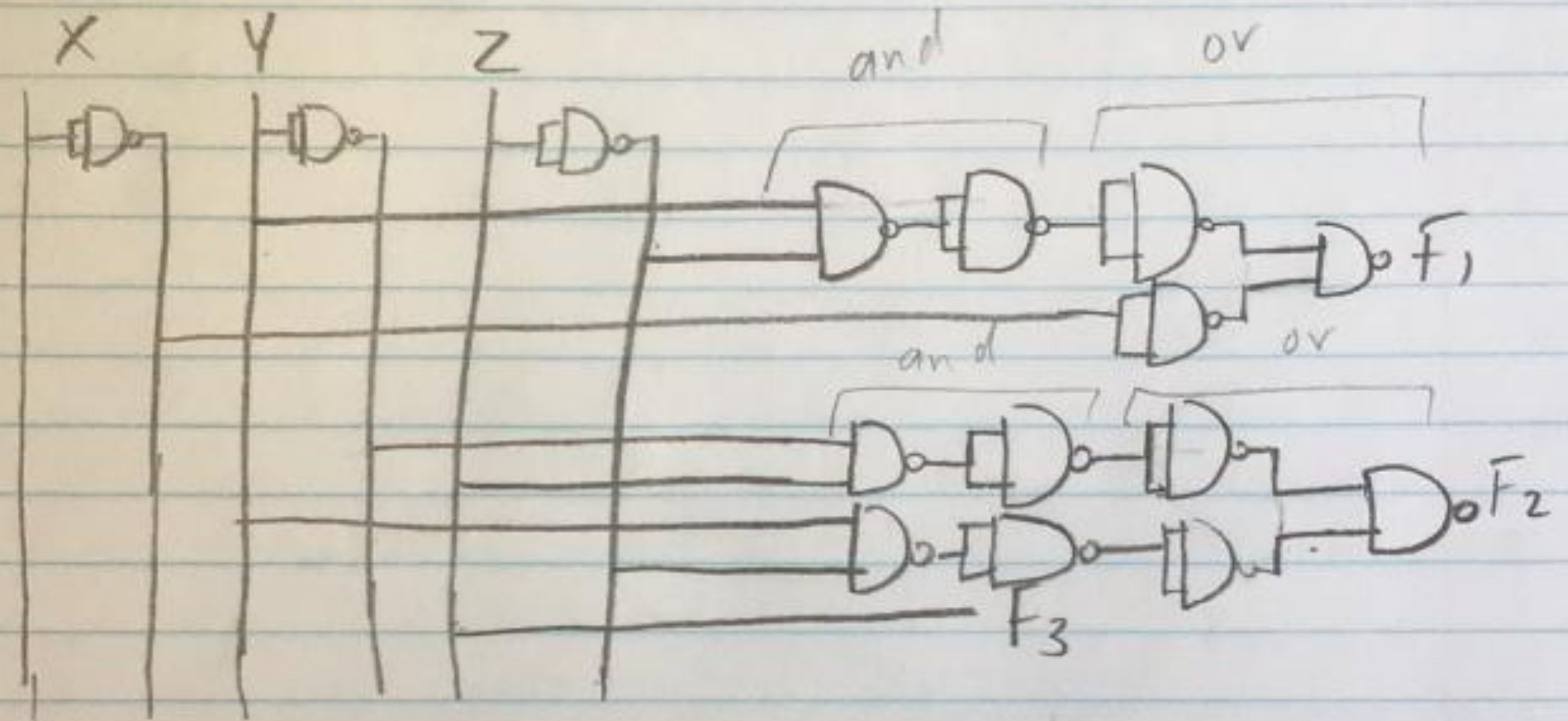
[› See more product details](#)



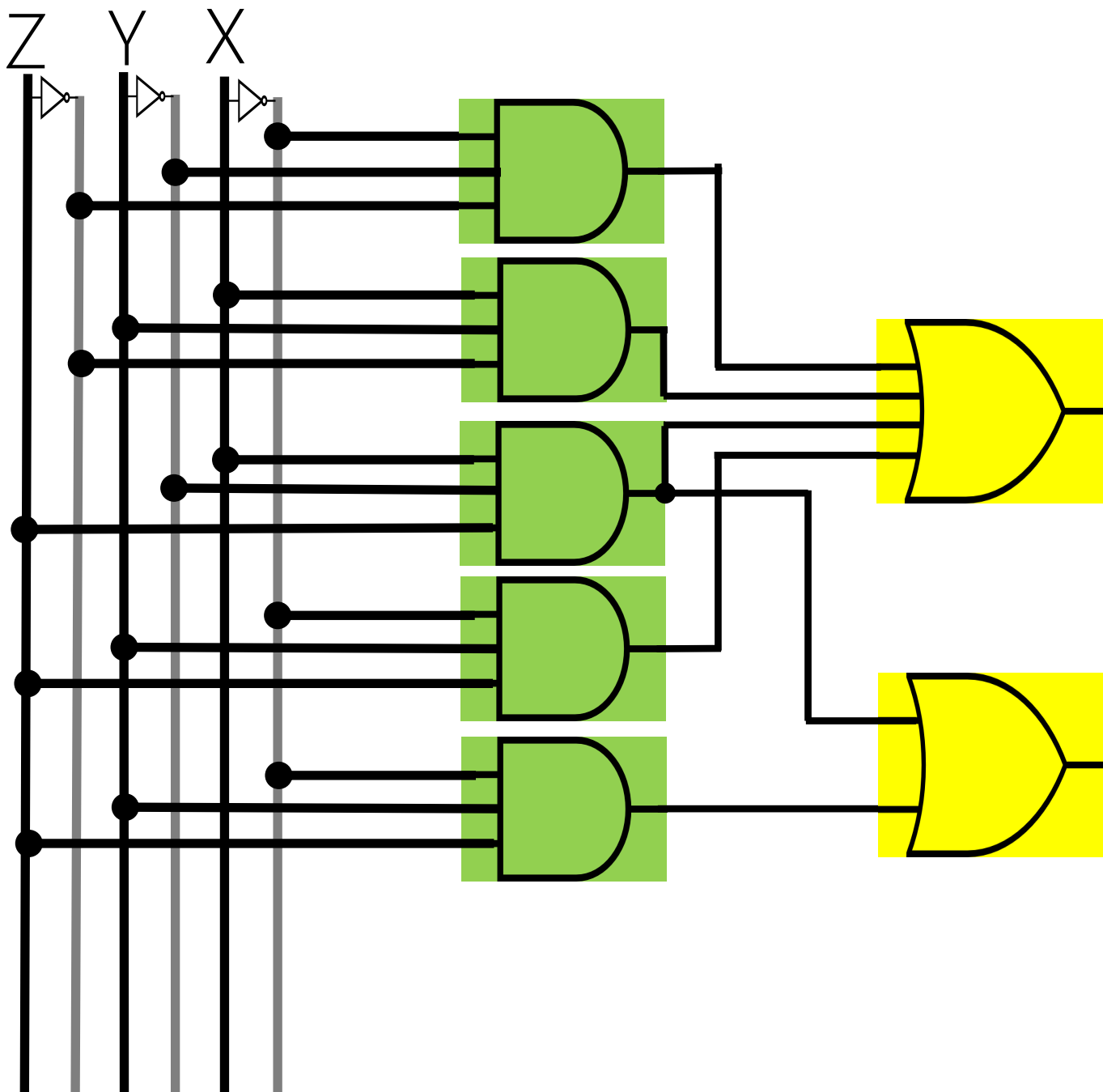
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UNIVERSAL GATE

SoP $\rightarrow$ {NAND}

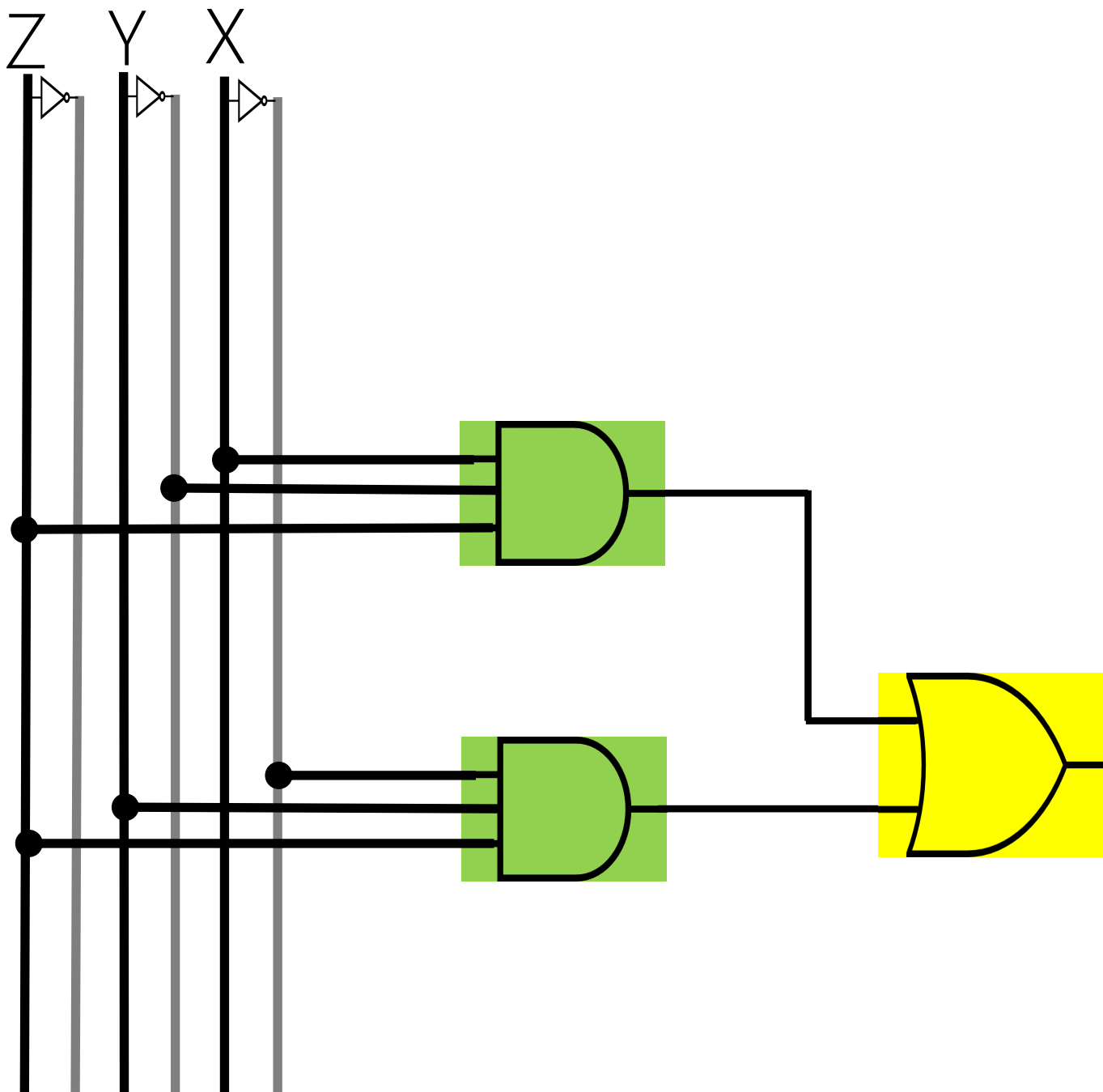


Credit goes to Haiqa Arain (Winter 2021)

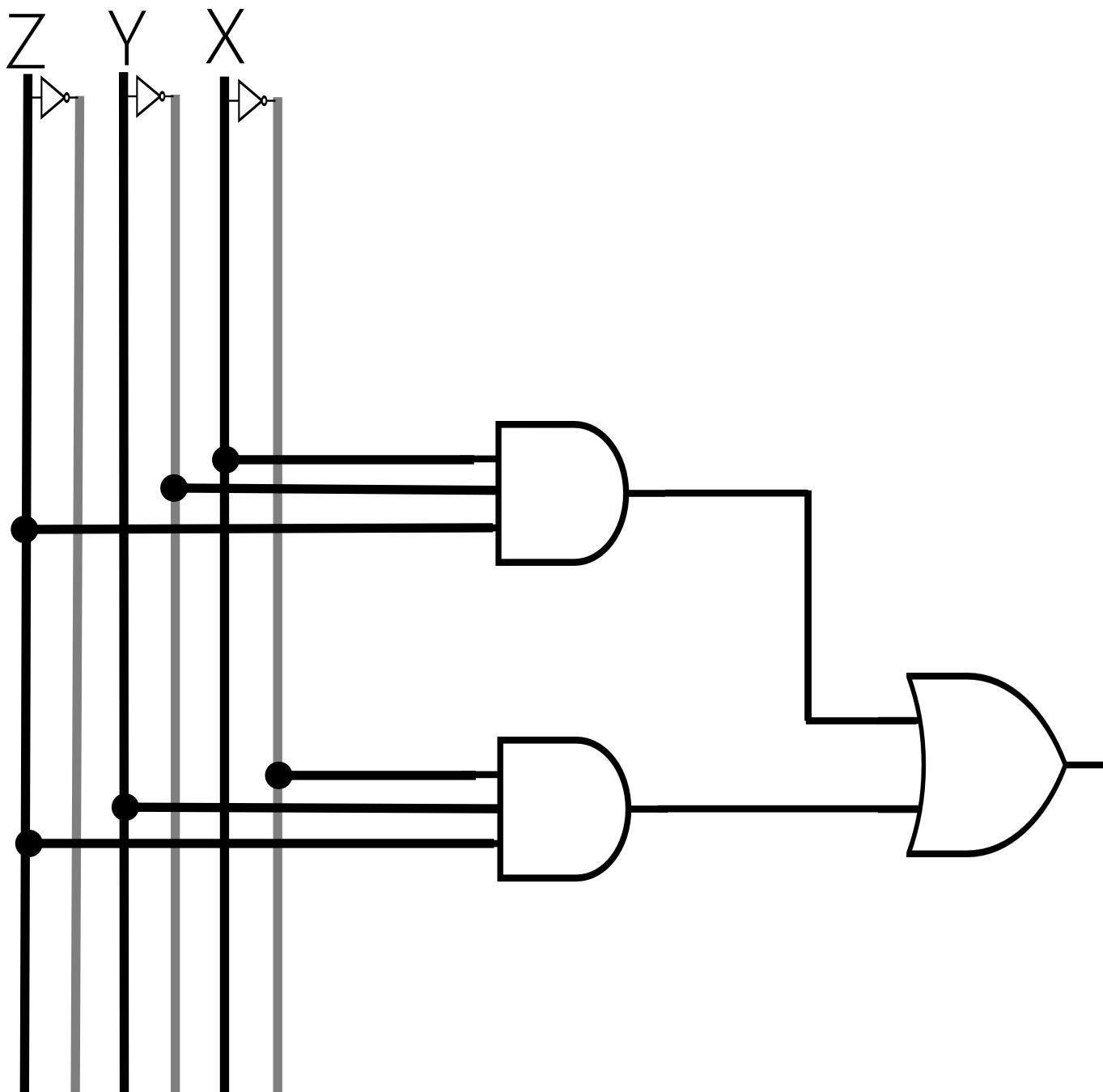


$$F_1 = Z'Y'X' + Z'YX + ZY'X + ZYX'$$

$$F_2 = ZY'X' + ZY'X$$

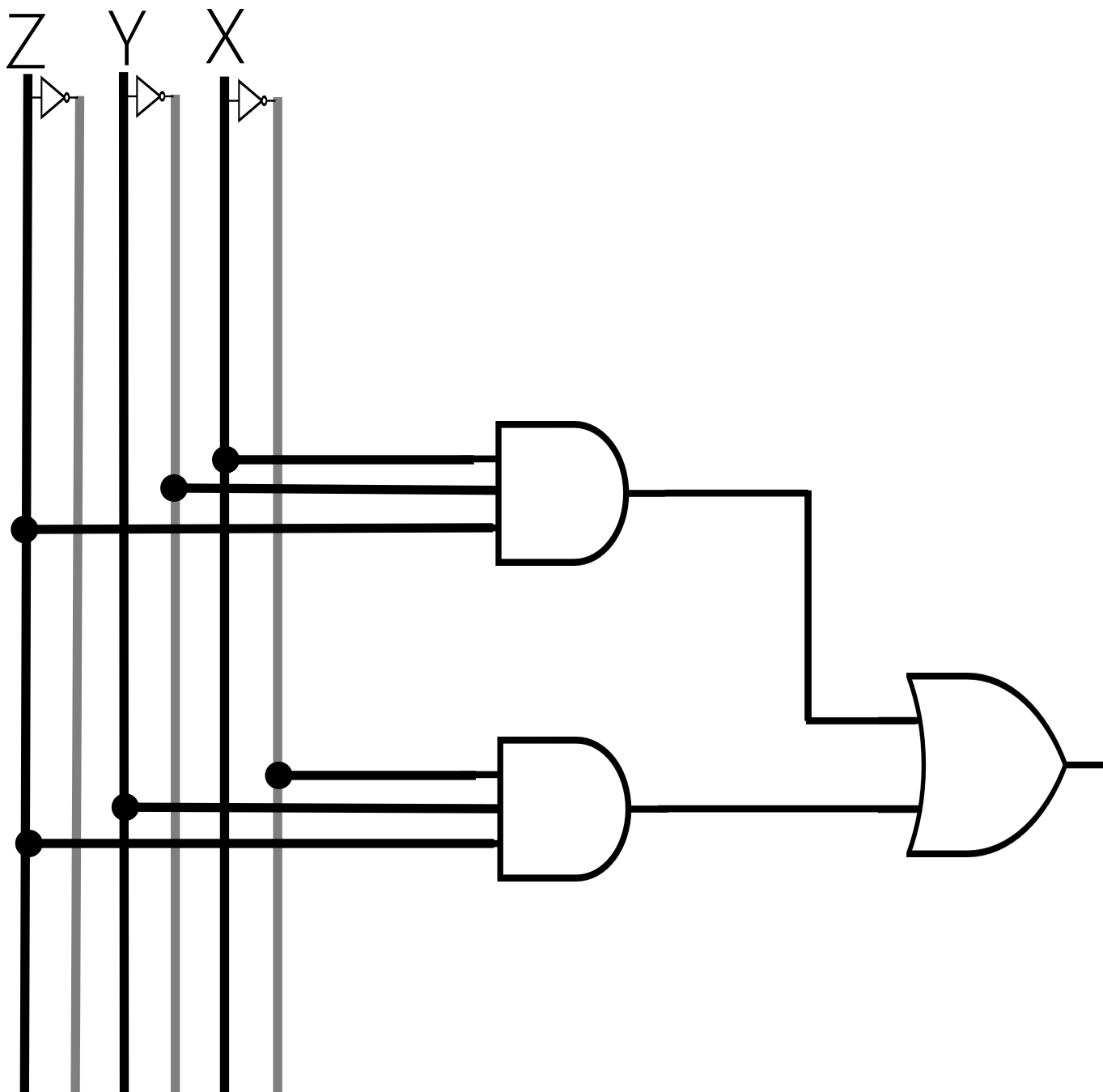


$$F_2 = ZY'X' + ZY'X$$

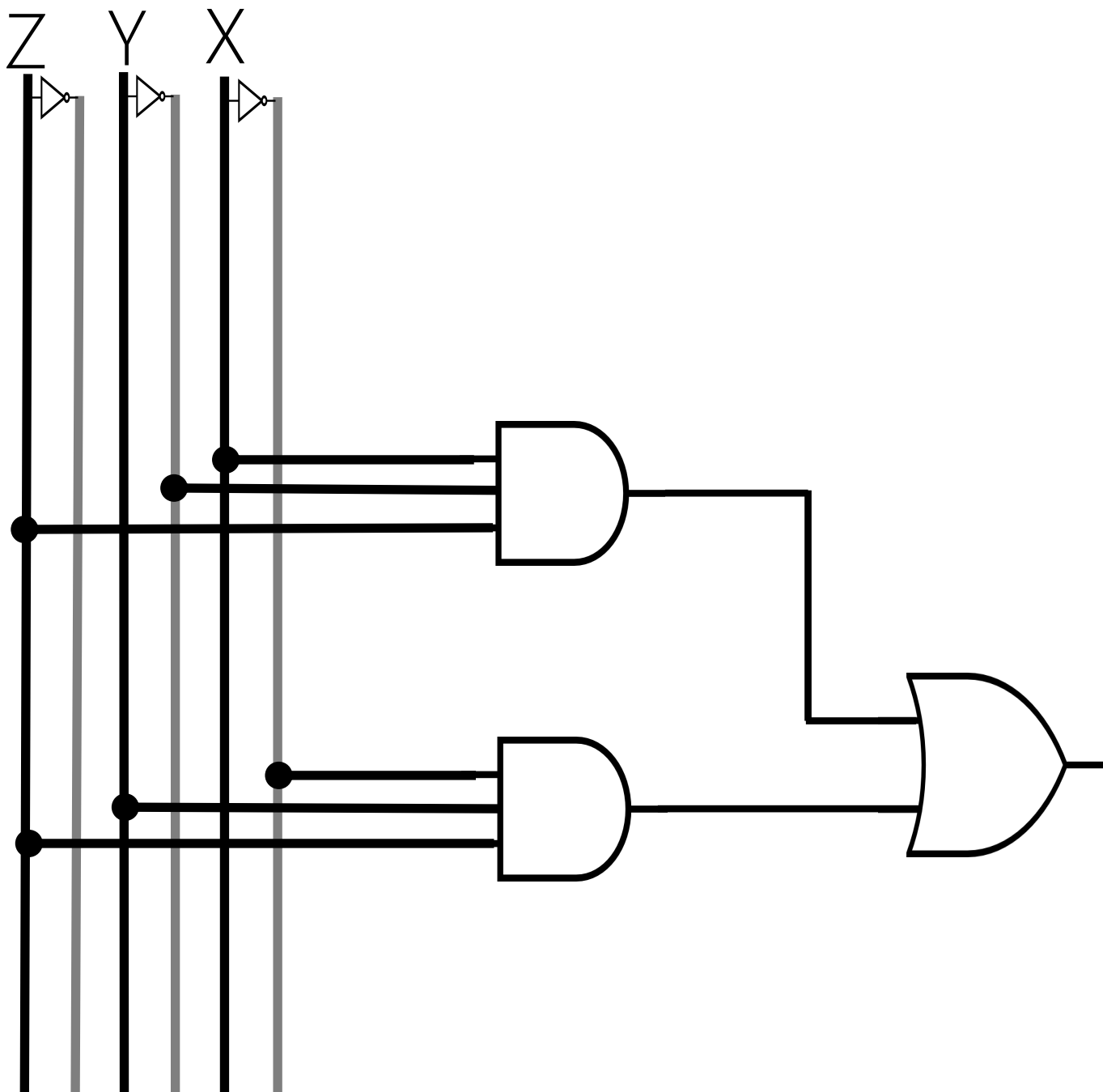


$$F_2 = m_4 + m_5$$

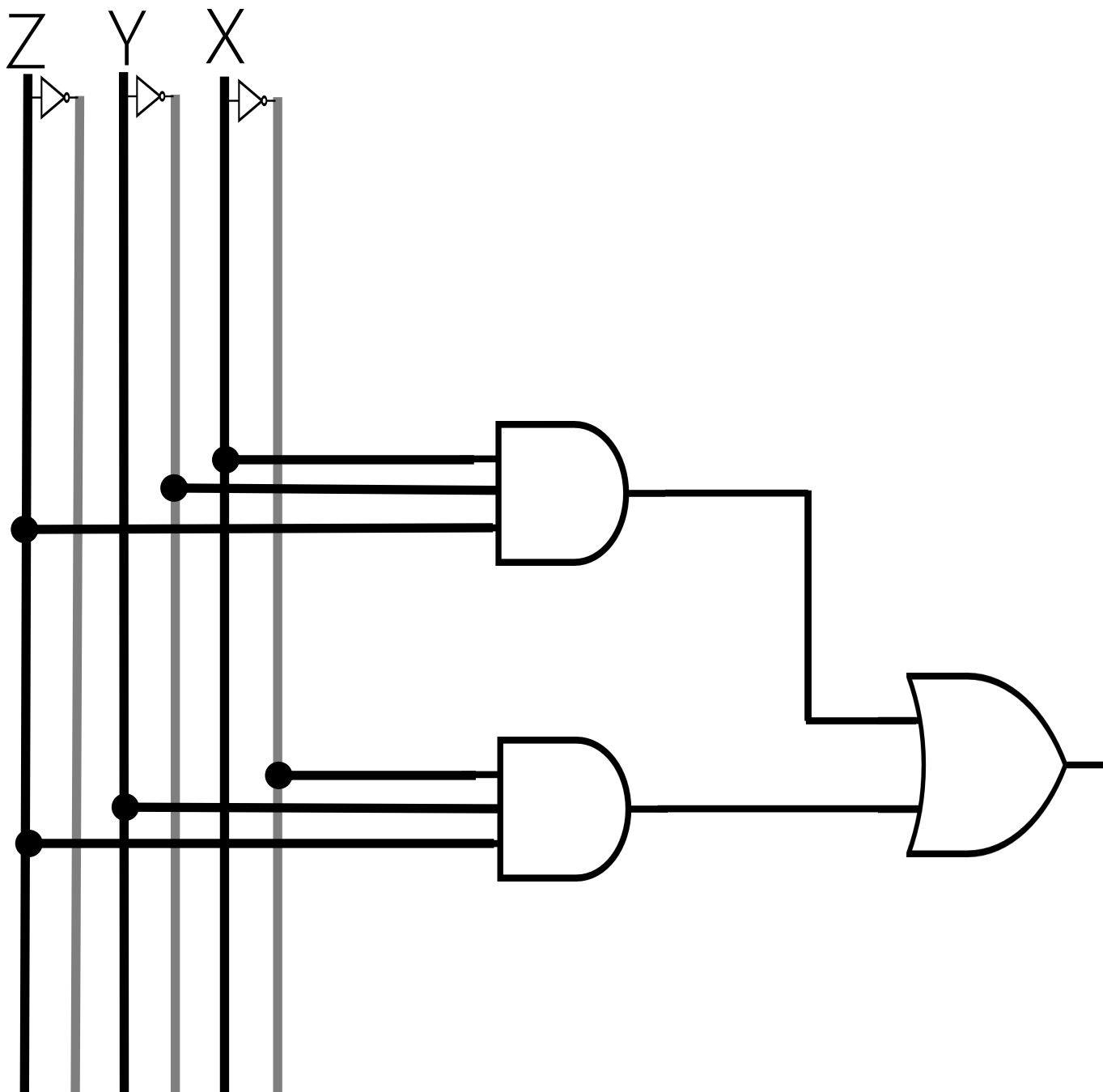
$$= ((F_2)')'$$



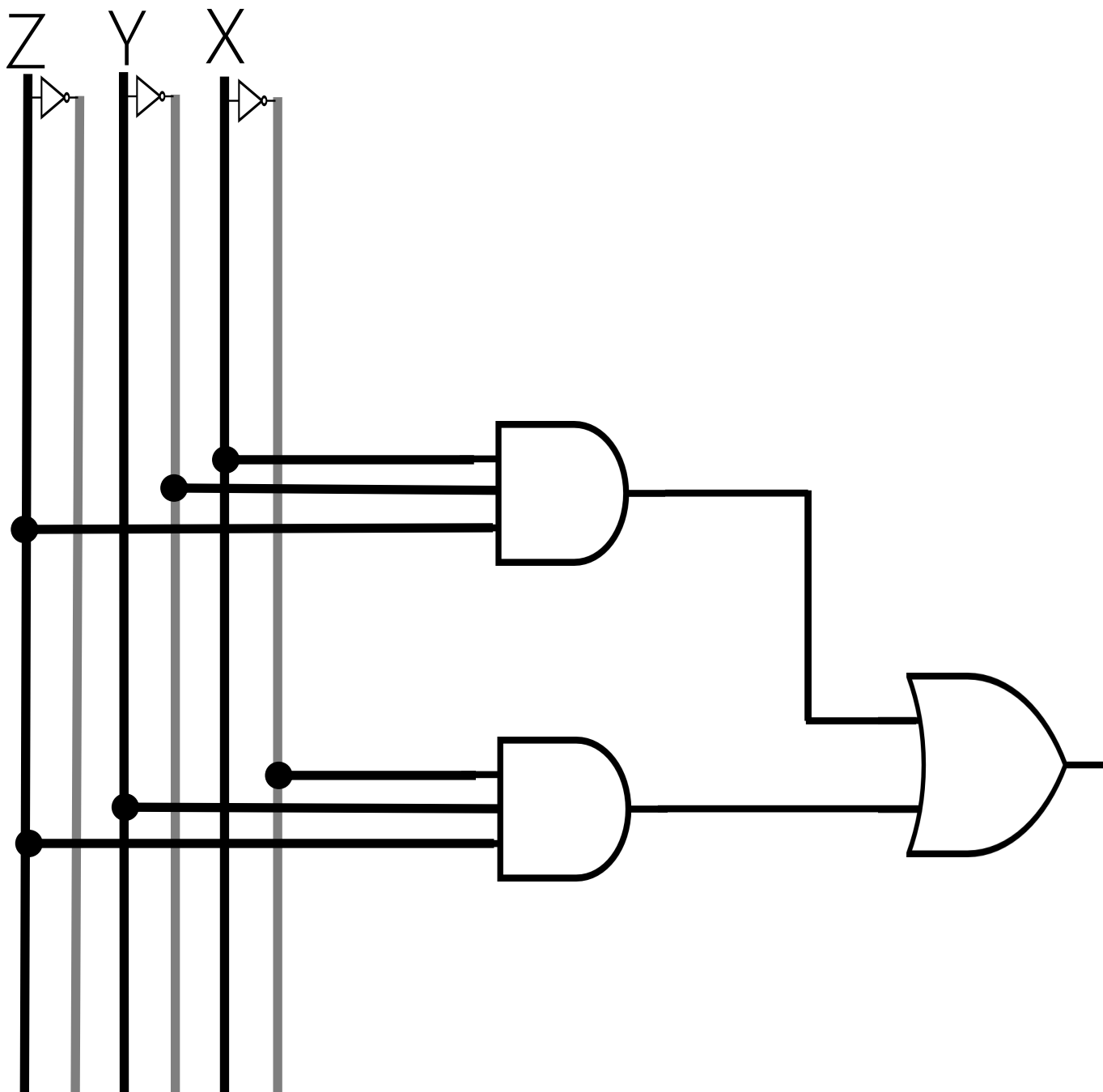
$$\begin{aligned}
 F_2 &= m_4 + m_5 \\
 &= ((F_2)')' \\
 &= ((m_4 + m_5)')'
 \end{aligned}$$



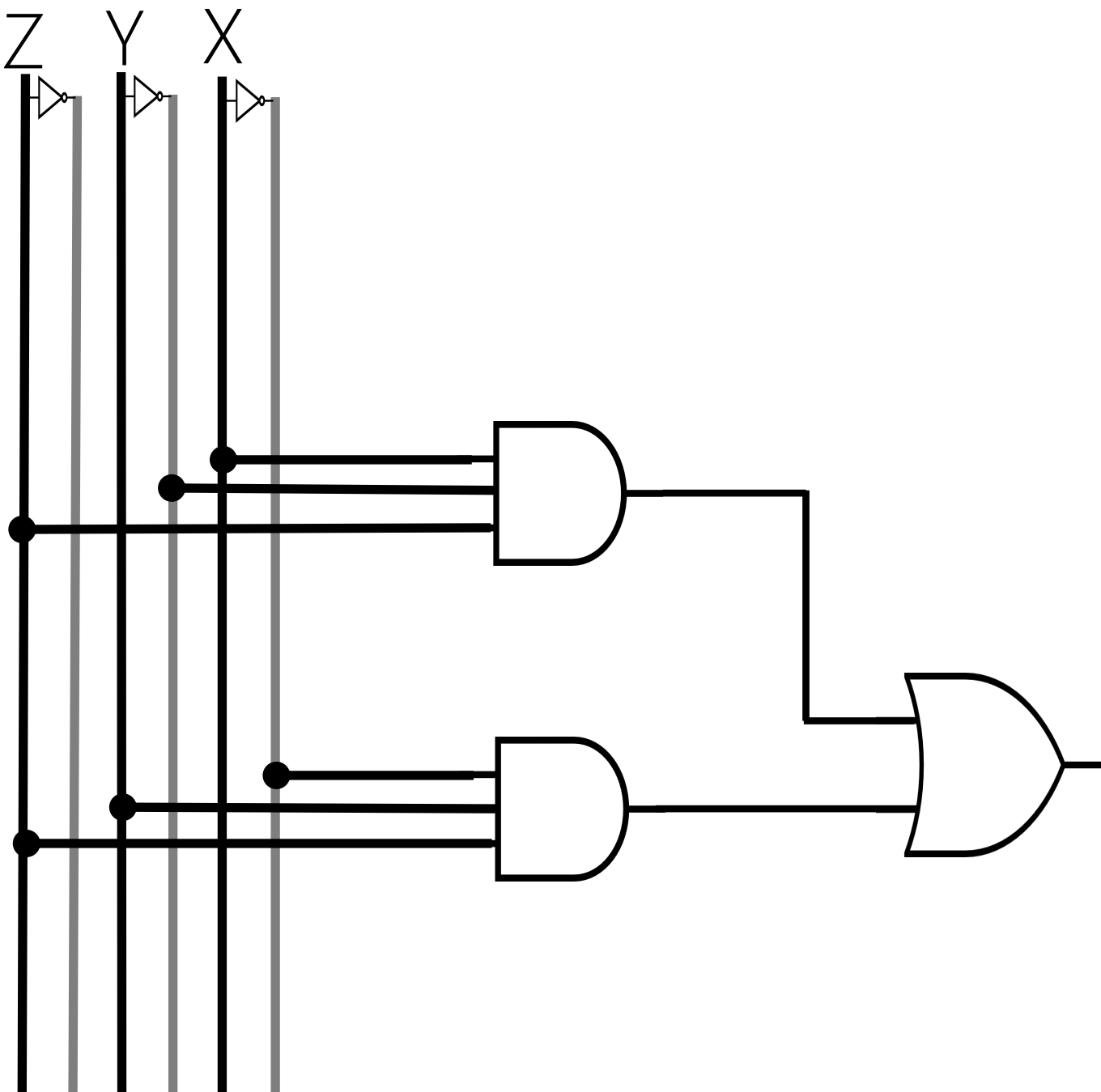
$$\begin{aligned}
 F_2 &= m_4 + m_5 \\
 &= ((F_2)')' \\
 &= ((m_4 + m_5)')' \\
 &= (m_4' m_5')'
 \end{aligned}$$



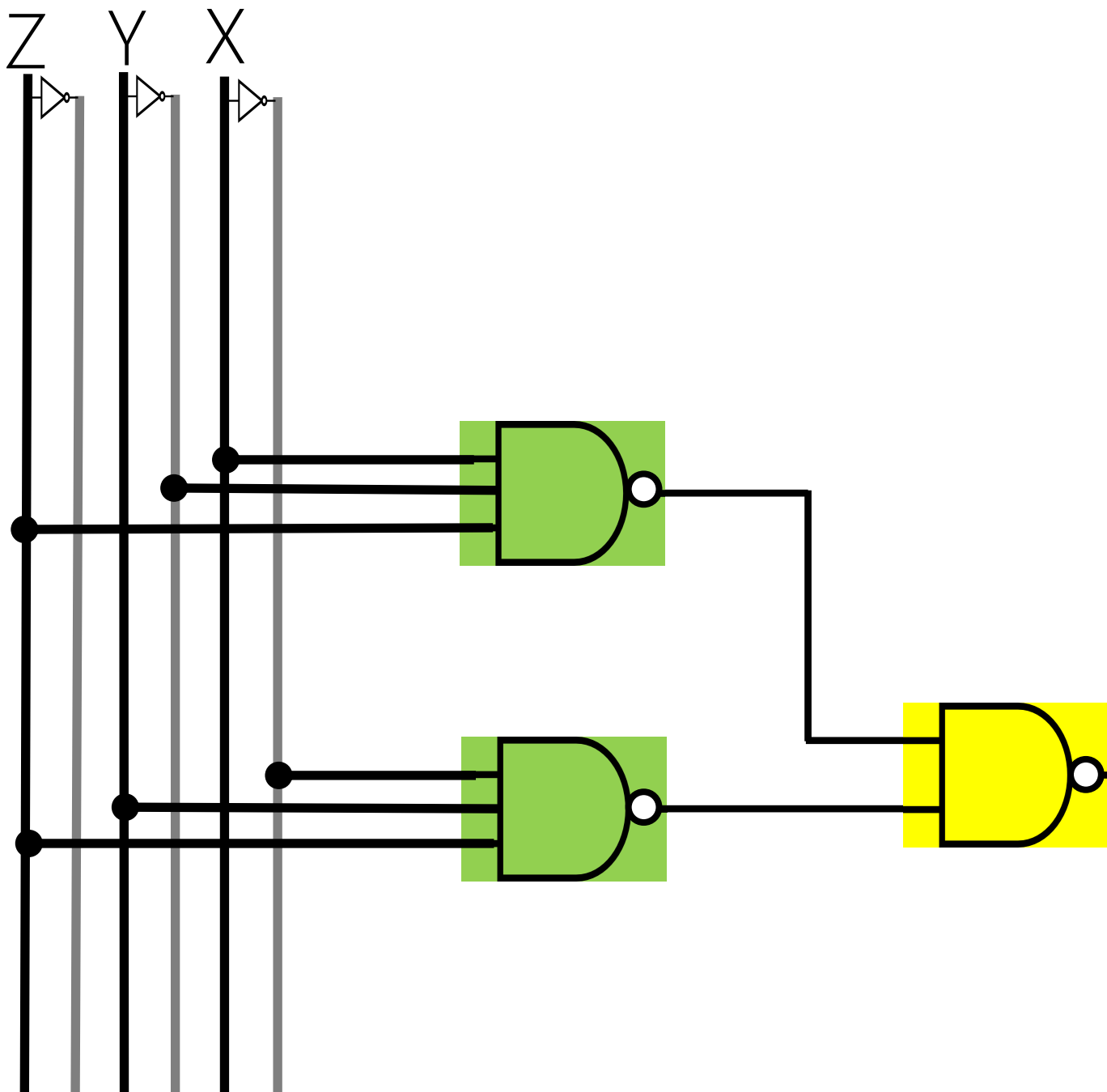
$$\begin{aligned}
 F_2 &= m_4 + m_5 \\
 &= ((F_2)')' \\
 &= ((m_4 + m_5)')' \\
 &= (m'_4 m'_5)' \\
 &= ((ZY'X')' (ZY'X)')'
 \end{aligned}$$



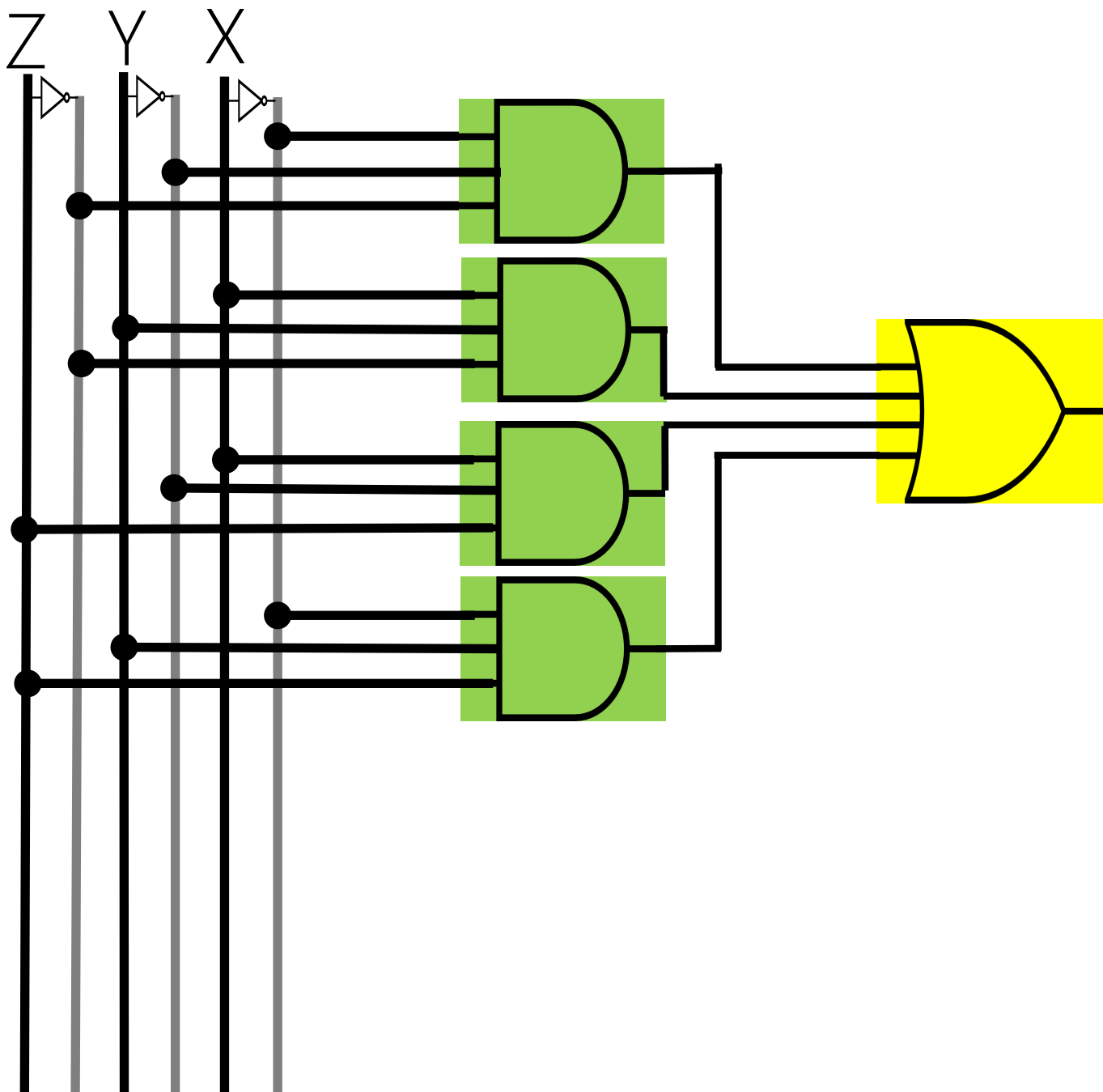
$$\begin{aligned}
 F_2 &= m_4 + m_5 \\
 &= ((F_2)')' \\
 &= ((m_4 + m_5)')' \\
 &= (m'_4 m'_5)' \\
 &= ((ZY'X')' (ZY'X)')' \\
 &= ((Z\uparrow Y'\uparrow X') (Z\uparrow Y'\uparrow X))'
 \end{aligned}$$



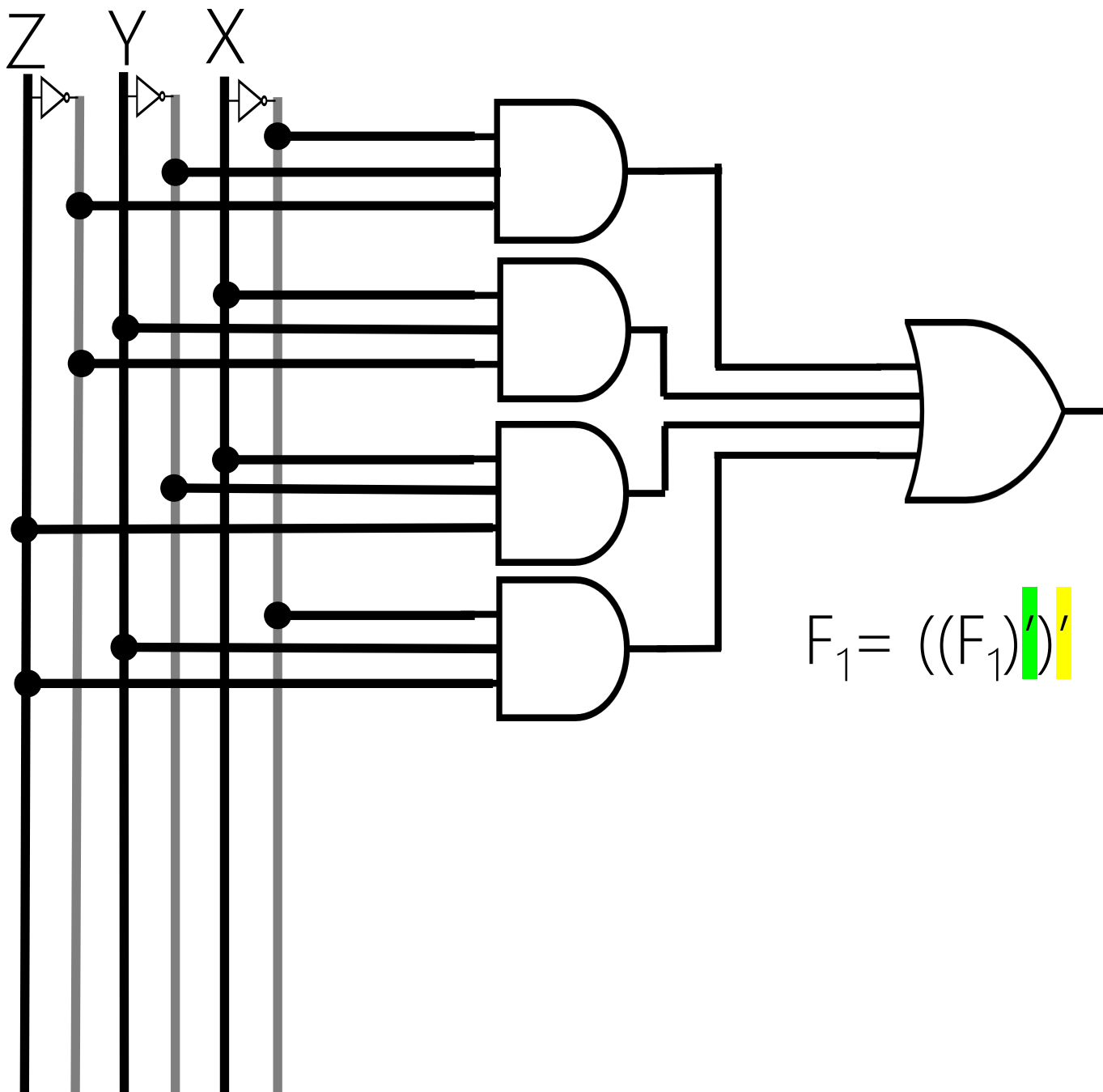
$$\begin{aligned}
 F_2 &= m_4 + m_5 \\
 &= ((F_2)')' \\
 &= ((m_4 + m_5)')' \\
 &= (m'_4 m'_5)' \\
 &= ((ZY'X')' (ZY'X)')' \\
 &= ((Z \uparrow Y' \uparrow X') (Z \uparrow Y' \uparrow X))' \\
 &= ((Z \uparrow Y' \uparrow X') \uparrow (Z \uparrow Y' \uparrow X))
 \end{aligned}$$



$$F_2 = ((Z \uparrow Y' \uparrow X') \uparrow (Z \uparrow Y' \uparrow X))$$

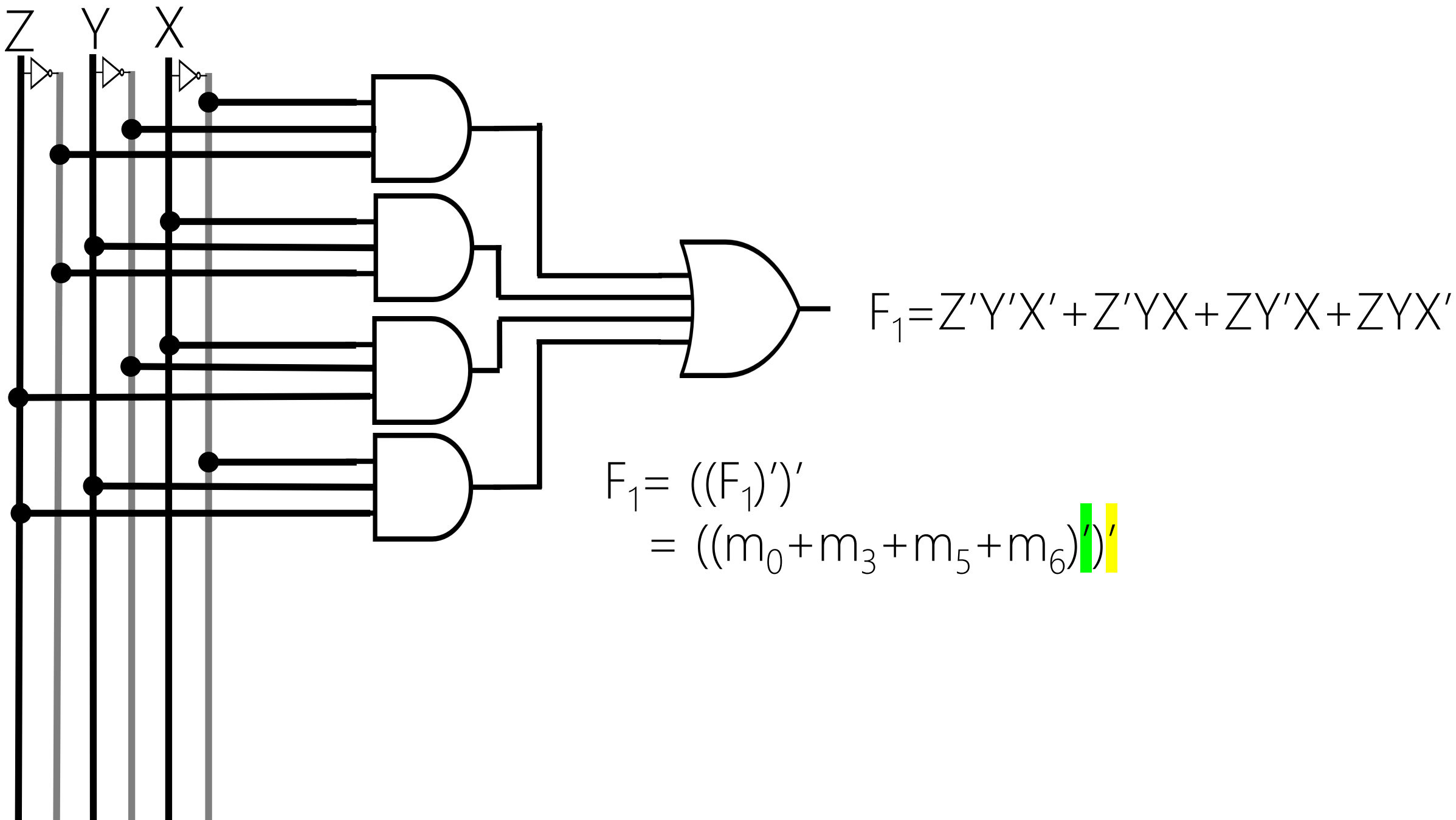


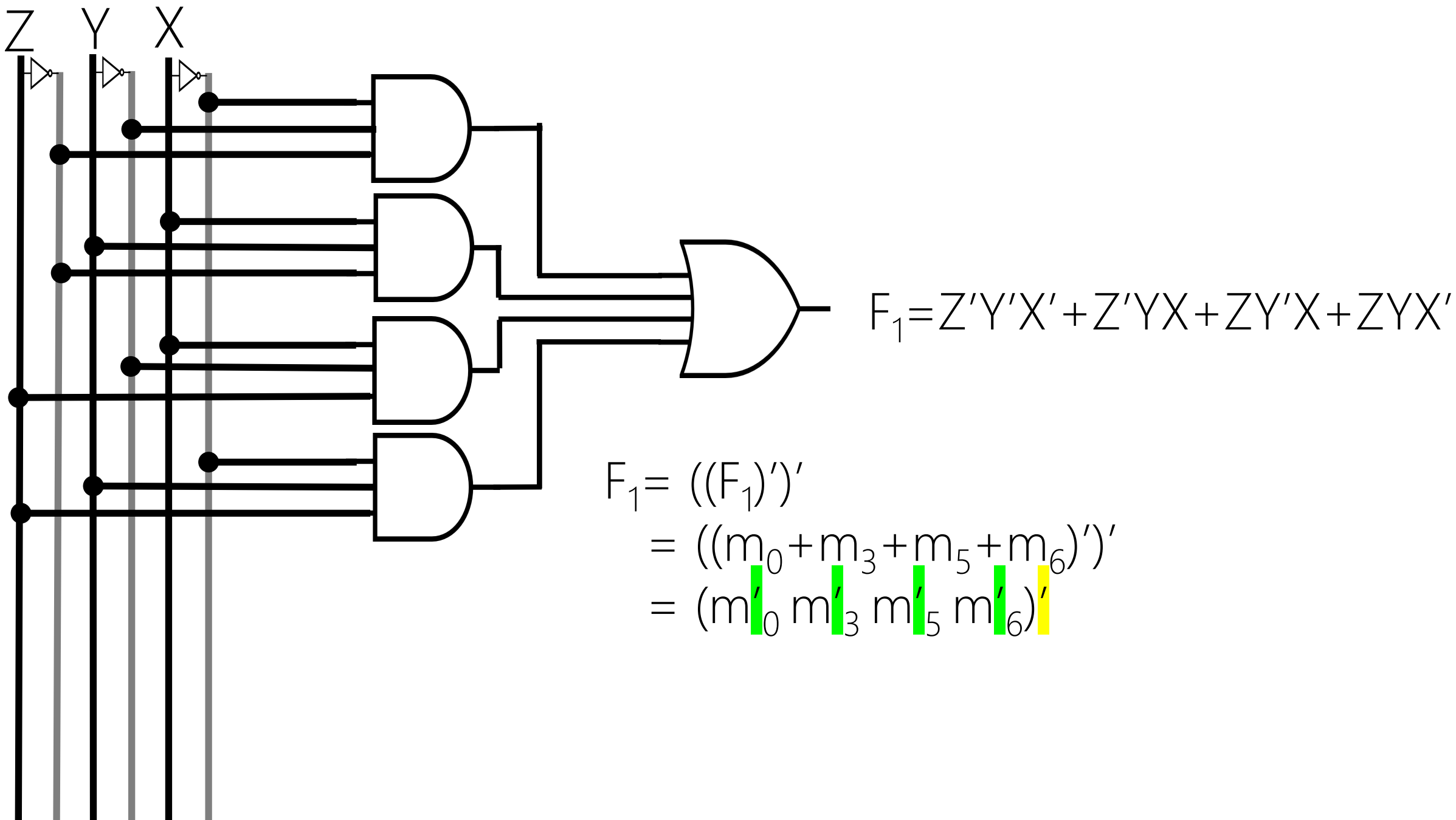
$$F_1 = Z'Y'X' + Z'YX + ZY'X + ZYX'$$



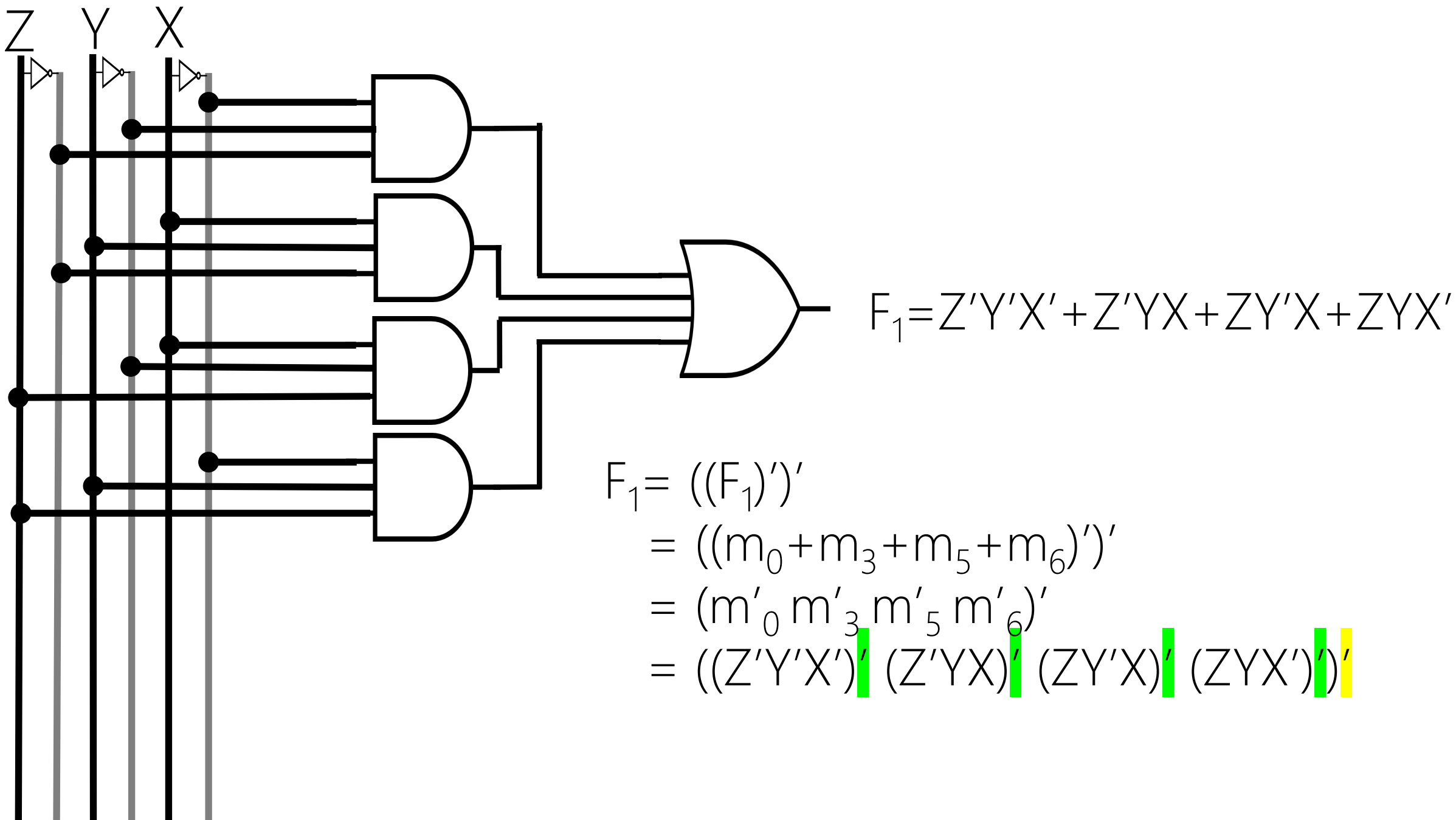
$$F_1 = Z'Y'X' + Z'YX + ZY'X + ZYX'$$

$$F_1 = ((F_1)')'$$

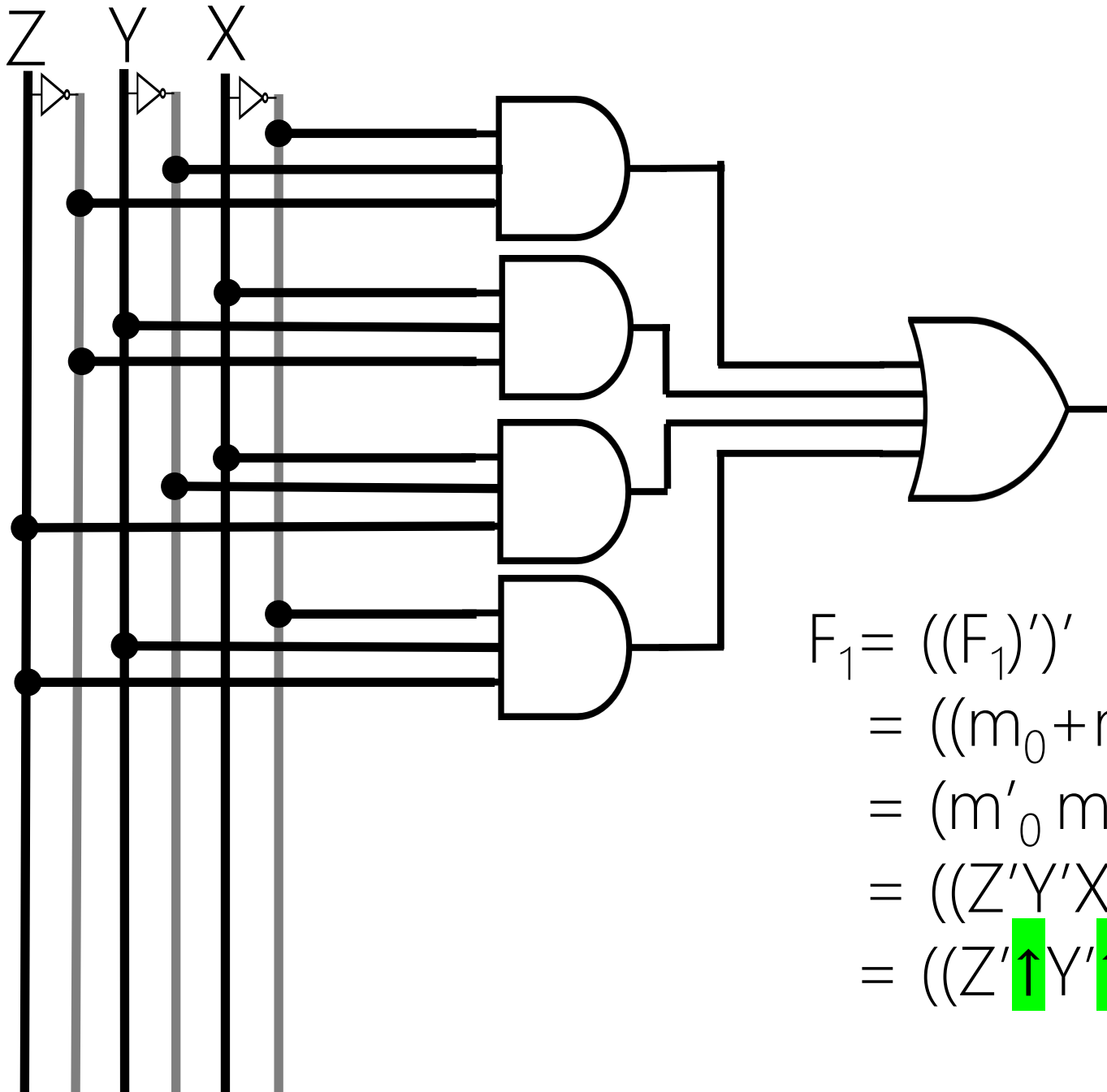




$$\begin{aligned}
 F_1 &= ((F_1)')' \\
 &= ((m_0 + m_3 + m_5 + m_6)')' \\
 &= (m'_0 m'_3 m'_5 m'_6)'
 \end{aligned}$$

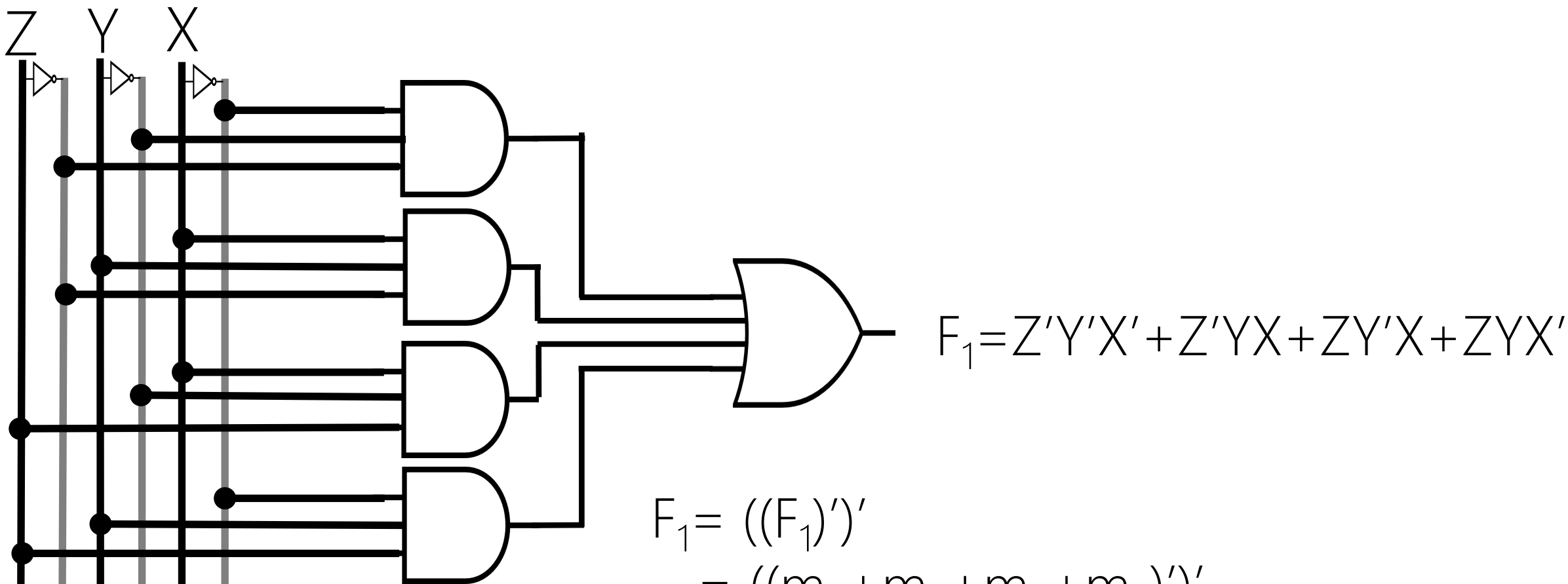


$$\begin{aligned}
 F_1 &= ((F_1)')' \\
 &= ((m_0 + m_3 + m_5 + m_6)')' \\
 &= (m'_0 m'_3 m'_5 m'_6)' \\
 &= ((Z'Y'X')' (Z'YX)' (ZY'X)' (ZYX')')'
 \end{aligned}$$



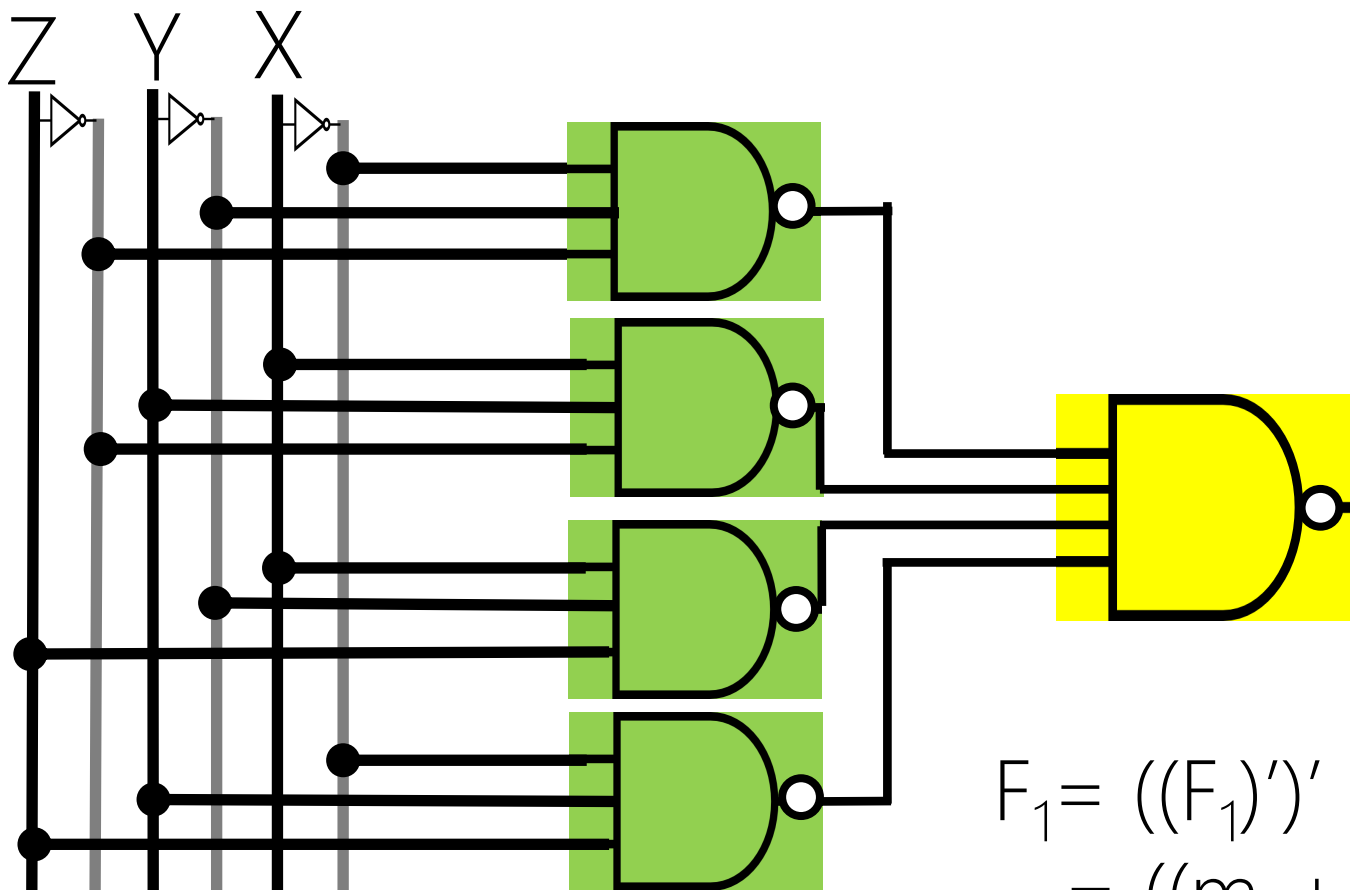
$$F_1 = Z'Y'X' + Z'YX + ZY'X + ZYX'$$

$$\begin{aligned}
 F_1 &= ((F_1)')' \\
 &= ((m_0 + m_3 + m_5 + m_6)')' \\
 &= (m'_0 m'_3 m'_5 m'_6)' \\
 &= ((Z'Y'X')' (Z'YX)' (ZY'X)' (ZYX')')' \\
 &= ((Z' \uparrow Y' \uparrow X') (Z' \uparrow Y \uparrow X) (Z \uparrow Y' \uparrow X) (Z \uparrow Y \uparrow X'))'
 \end{aligned}$$



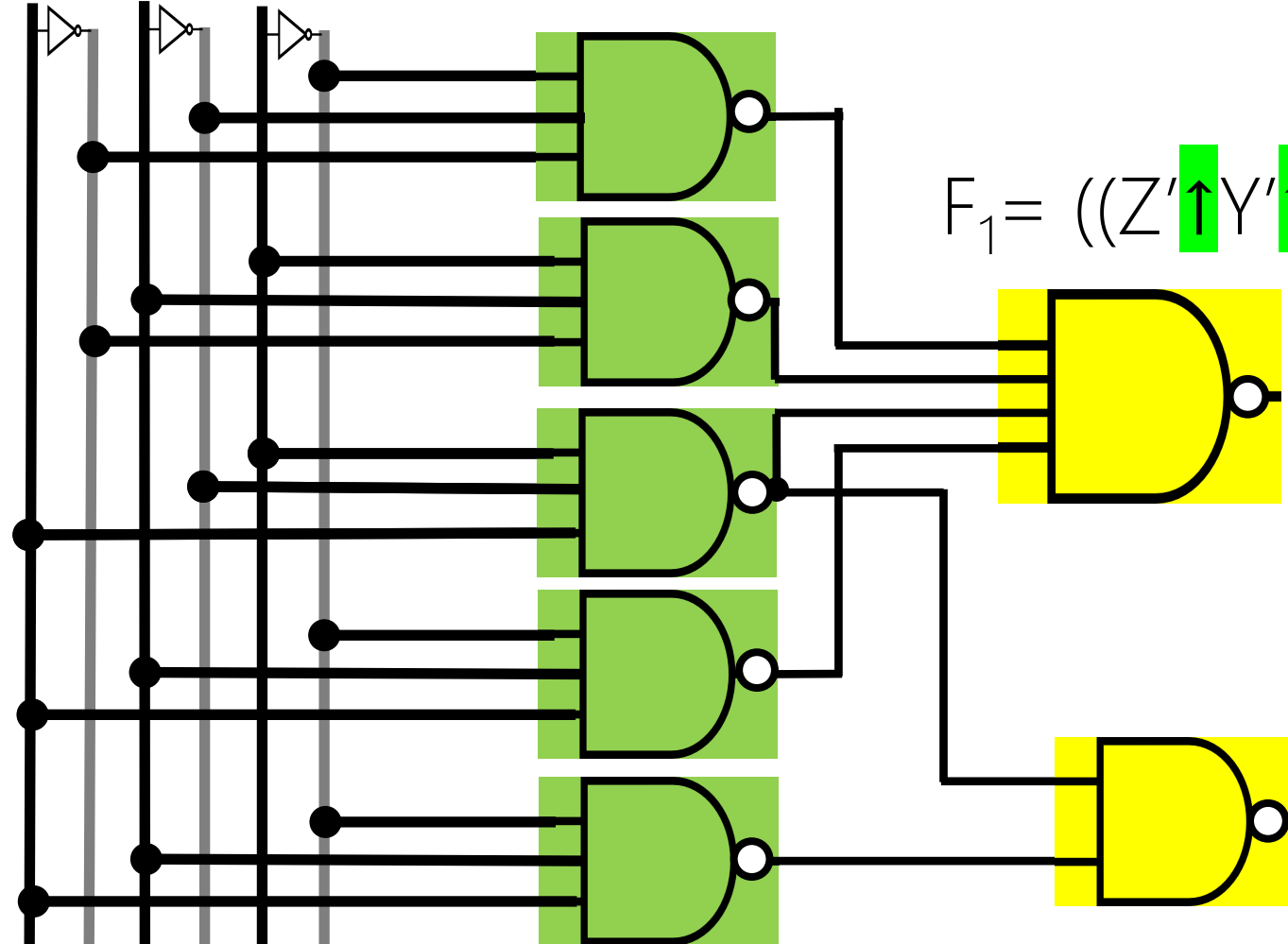
$$F_1 = Z'Y'X' + Z'YX + ZY'X + ZYX'$$

$$\begin{aligned}
 F_1 &= ((F_1)')' \\
 &= ((m_0 + m_3 + m_5 + m_6)')' \\
 &= (m'_0 m'_3 m'_5 m'_6)' \\
 &= ((Z'Y'X')' (Z'YX)' (ZY'X)' (ZYX')')' \\
 &= ((Z' \uparrow Y' \uparrow X') (Z' \uparrow Y \uparrow X) (Z \uparrow Y' \uparrow X) (Z \uparrow Y \uparrow X'))' \\
 &= ((Z' \uparrow Y' \uparrow X') \uparrow (Z' \uparrow Y \uparrow X) \uparrow (Z \uparrow Y' \uparrow X) \uparrow (Z \uparrow Y \uparrow X'))
 \end{aligned}$$



$$\begin{aligned}
 F_1 &= ((F_1)')' \\
 &= ((m_0 + m_3 + m_5 + m_6)')' \\
 &= (m'_0 m'_3 m'_5 m'_6)' \\
 &= ((Z'Y'X')' (Z'YX)' (ZY'X)' (ZYX')')' \\
 &= ((Z' \uparrow Y' \uparrow X') (Z' \uparrow Y \uparrow X) (Z \uparrow Y' \uparrow X) (Z \uparrow Y \uparrow X'))' \\
 &= ((Z' \uparrow Y' \uparrow X') \uparrow (Z' \uparrow Y \uparrow X) \uparrow (Z \uparrow Y' \uparrow X) \uparrow (Z \uparrow Y \uparrow X'))
 \end{aligned}$$

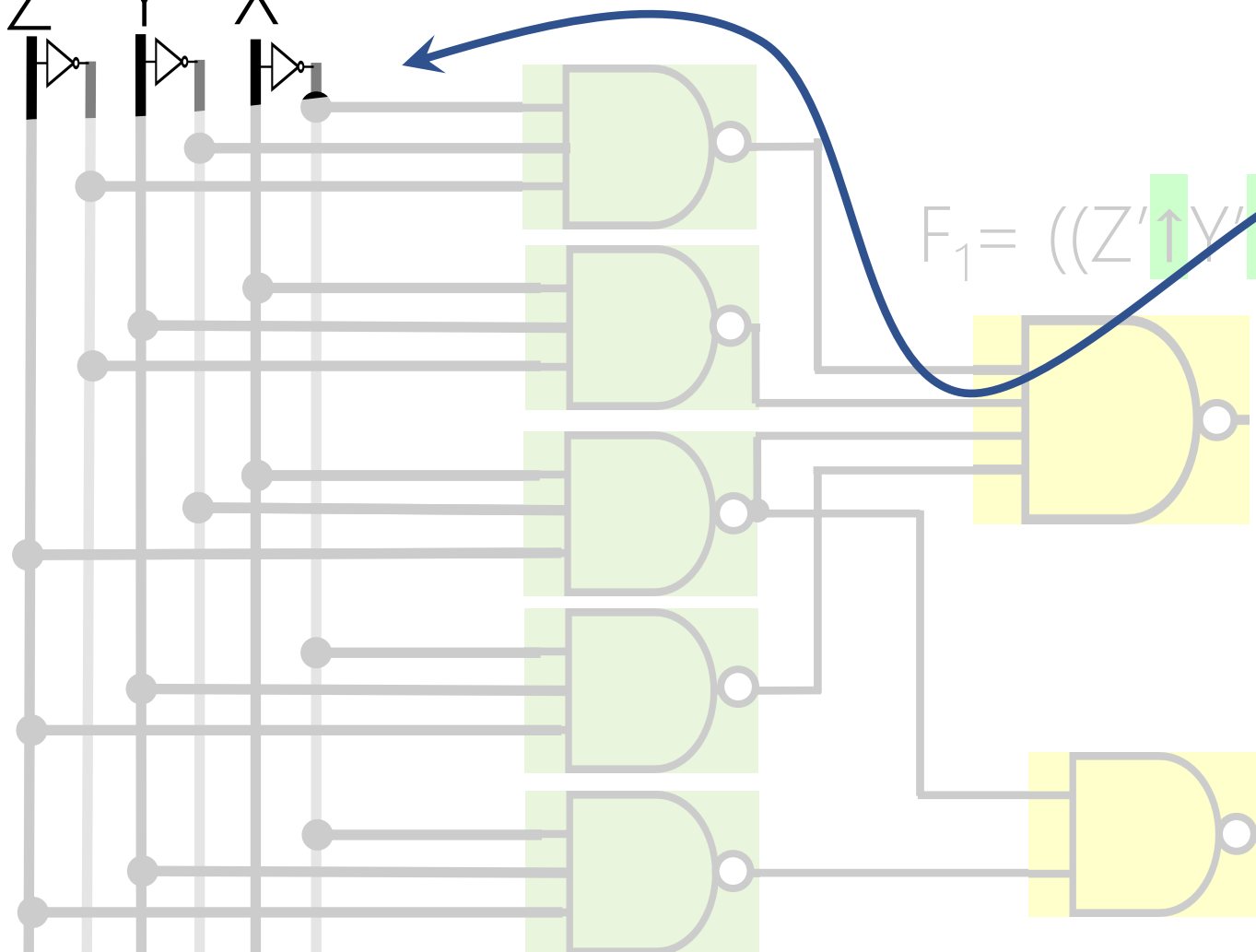
Z Y X



$$F_1 = ((Z' \uparrow Y' \uparrow X') \uparrow (Z' \uparrow Y \uparrow X) \uparrow (Z \uparrow Y' \uparrow X) \uparrow (Z \uparrow Y \uparrow X'))$$

$$F_2 = ((Z \uparrow Y' \uparrow X') \uparrow (Z \uparrow Y' \uparrow X))$$

Z Y X

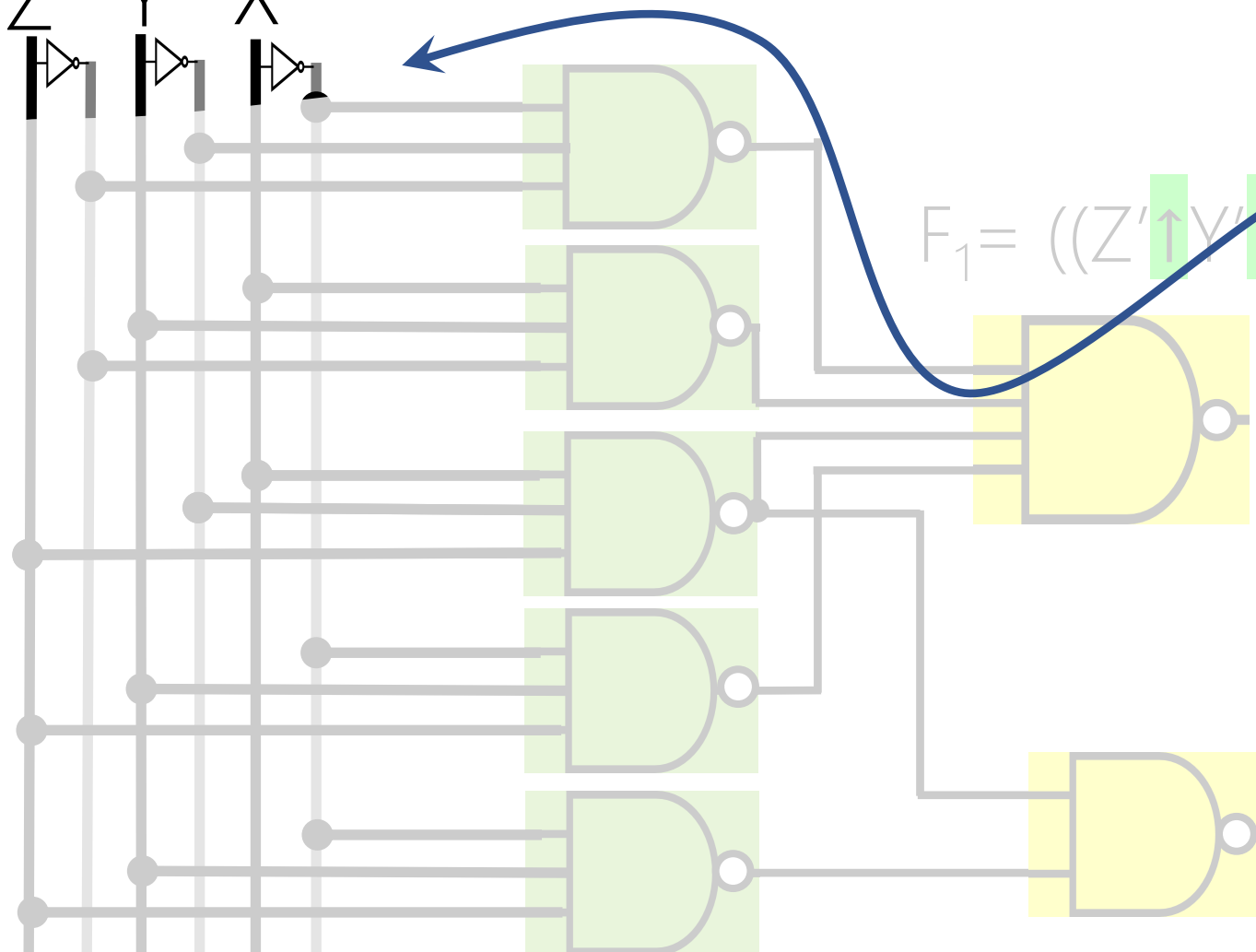


$$F_1 = ((Z' \uparrow Y' \uparrow X') \uparrow (Z' \uparrow Y \uparrow X) \uparrow (Z \uparrow Y' \uparrow X) \uparrow (Z \uparrow Y \uparrow X'))$$

NOT by NAND?

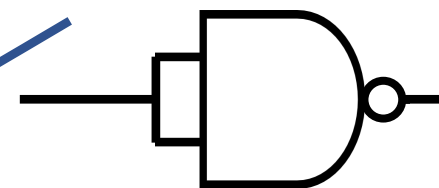
$$F_2 = ((Z \uparrow Y' \uparrow X') \uparrow (Z \uparrow Y' \uparrow X))$$

Z Y X



$$F_1 = ((Z' \uparrow Y' \uparrow X') \uparrow (Z' \uparrow Y \uparrow X) \uparrow (Z \uparrow Y' \uparrow X) \uparrow (Z \uparrow Y \uparrow X'))$$

NOT by NAND?



$$F_2 = ((Z \uparrow Y' \uparrow X') \uparrow (Z \uparrow Y' \uparrow X))$$

UNIVERSAL GATE {NAND}

$$F = (F')' = (\text{SoP}')'$$

UNIVERSAL GATE

{NOR}

UNIVERSAL GATE

$\text{PoS} \rightarrow \{\text{NOR}\}$

$$F_{\text{PoS}} = (F')'$$

Lecture Assignment

RECAP

Any Boolean Function F:

- Sum (OR) of Products (ANDs)
- Sum of **minterms** for Entries with **1**
 - ANDs-OR
 - NAND via $(F')'$
- Product (AND) of Sums (ORs)
- Product of **MAXTERMS (NOT minterms)** for Entries with **0**
 - ORs-AND
 - NOR via $(F')'$

UNIVERSAL GATE

SoP $\rightarrow$ {NAND} ✓

SoP $\rightarrow$ {NOR} ?

PoS $\rightarrow$ {NOR} ✓

PoS $\rightarrow$ {NAND} ?

Lecture Assignment